

CHEM 3110: DESCRIPTIVE INORGANIC CHEMISTRY



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CHEM 3110 - Descriptive Inorganic Chemistry

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TABLE OF CONTENTS

Licensing

1: Introduction to Inorganic Chemistry

- 1.1: What is Inorganic Chemistry?
- 1.2: Inorganic vs Organic Chemistry
- 1.3: History of Inorganic Chemistry

2: Atomic Theory

- 2.1: Quantum Numbers and Atomic Wavefunctions
- 2.2: Shielding
- 2.3: Aufbau Principle
- 2.4: Periodic Properties of Atoms

3: Bonding Theories

- 3.1: Types of Bonding
- 3.2: Electronegativity Values
 - 3.2A: Pauling Electronegativity Values
 - 3.2B: Mulliken Electronegativity Values
 - 3.2C: Allred-Rochow Electronegativity Values
- 3.3: Lewis Electron-Dot Diagrams
 - 3.3A: Formal Charge
 - 3.3B: Resonance
 - 3.3C: Expanded Octets
 - 3.3D: Limitation of Lewis Theory
- 3.4: Valence-Shell Electron-Repulsion Theory
- 3.5: Molecular Polarity
- 3.6: Valence Bond Theory
- 3.7: Molecular Orbital Theory
 - 3.7A: Orbital Overlap
 - 3.7B: Homonuclear Diatomic Molecules
 - 3.7C: Heteronuclear Diatomic Molecules

4: Molecular Symmetry and Point Groups

- 4.1: Symmetry Operations and Elements
- 4.2: Point Groups
- 4.3: Introduction to Character Tables
- 4.4: Applications of Symmetry in Chemistry

5: Structure and Energetics of Solids

- 5.1: Crystal Structures and Unit Cells
- 5.2: Energetics of Ionic Solids- Lattice Energy
- 5.3: Lattice Enthalpies and Born Haber Cycles
- 5.4: Lattice Energy and Solubility
- 5.5: Bonding in Metals and Semiconductors
- 5.6: Nanomaterials

6: Acid-Base and Donor-Acceptor Chemistry

- 6.1: Arrhenius Model
- 6.2: Brønsted-Lowry Model
- 6.3: Lewis Concept and Frontier Orbitals
- 6.4: Hard and Soft Acids and Bases

7: Reduction and Oxidation Chemistry

- 7.1: Balancing Redox Reactions
- 7.2: Electrochemical Potentials
- 7.3: Latimer Diagrams
- 7.4: Frost Diagrams
- 7.5: Pourbaix Diagrams

8: Coordination Chemistry- Structure and Isomers

- 8.1: History
- 8.2: Nomenclature and Ligands
- 8.3: Coordination Numbers and Structures
- 8.4: Isomerism

9: Coordination Chemistry- Bonding

- 9.1: Crystal Field Theory
- 9.2: Crystal Field Stabilization Energy
- 9.3: Jahn-Teller Distortions
- 9.4: Factors That Affect Crystal Field Splitting
- 9.5: Ligand Field Theory

10: Coordination Chemistry- Reactions and Mechanisms

- 10.1: Thermodynamic Stability of Metal Complexes
- 10.2: Trends in Kinetic Lability
- 10.3: Ligand Substitution Mechanisms
- 10.4: The Trans Effect
- 10.5: Electron Transfer Reactions

11: Organometallic Chemistry

- 11.1: Organometallic Ligands
- 11.2: The 18 Electron Rule
- 11.3: Oxidative Addition
- 11.4: Reductive Elimination
- 11.5: Migratory Insertion- 1,2-Insertions
- 11.6: β -Elimination Reactions
- 11.7: Organometallic Catalysts

12: Bioinorganic Chemistry

- 12.1: Biological Significance of Metals
- 12.2: Introduction to Amino Acids and Proteins
- 12.3: Biological Dioxygen Transport and Storage
- 12.4: Biological Metal Storage
- 12.5: Zinc as Lewis Acid and Template

- [12.6: Electron Transfer](#)

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

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CHAPTER OVERVIEW

1: Introduction to Inorganic Chemistry

This chapter will introduce the breadth of the field of inorganic chemistry

Learning Objectives

- Identify different subfields of inorganic chemistry
- Recognize the historical and modern distinction between the fields of inorganic and organic chemistry

[1.1: What is Inorganic Chemistry?](#)

[1.2: Inorganic vs Organic Chemistry](#)

[1.3: History of Inorganic Chemistry](#)

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1.1: What is Inorganic Chemistry?

Where did the name "Inorganic Chemistry" come from? Well, the term "Organic Chemistry" literally means the *chemistry of life*. Organic chemistry is the study of carbon-based molecules because the first molecules that were isolated from living organisms contained carbon. On the other hand, minerals and other non-living things seemed to be made of other elements. For some time in our history, scientists believed that the chemical difference between living and non-living things was carbon. So, if "organic" molecules are the molecules of life, then is "inorganic chemistry" the "chemistry of death"? Almost? "Inorganic" chemistry historically meant the chemistry of "non-living" things; and these were non-carbon based molecules and ions.

The names "organic" and "inorganic" come from science history, and still today a generally-accepted definition of *Inorganic Chemistry is the study of non-carbon molecules, or all the elements on the periodic table except carbon* (Figure 1.1.1). But, this definition is not completely correct because the field of Inorganic Chemistry also includes organometallic compounds and the study of some carbon-based molecules that have properties that are familiar to metals (like conduction of electricity). This makes the field of inorganic chemistry very broad, and practically limitless. A great way to understand the breadth of the field is to take a look at the abstracts in the latest article of Inorganic Chemistry. Or, check out the 20 most-read articles from this past year using the links below.

Inorganic (not life)

Life

* Lanthanide series

* Actinide series

Figure 1.1.1. An illustration of the historic meaning of "organic" and "inorganic". The modern understanding of organic and inorganic chemistry is not consistent with these historical meanings. (Kathryn Haas; CC-NC-BY-SA)

External links:

- The journal, Inorganic Chemistry: <https://pubs.acs.org/journal/inocaj>
- The latest issue of Inorganic chemistry: <https://pubs.acs.org/toc/inocaj/current>
- The most popular Inorganic Chemistry articles from the past month and the past year: <https://pubs.acs.org/action/showMostReadArticles?journalCode=inocaj>

Practice

? What are the Sub-Fields of Inorganic Chemistry?

To appreciate the breadth of Inorganic Chemistry, go to the most recent issue of Inorganic Chemistry and look at the titles and visual abstracts. Identify at least 4 sub-fields of Inorganic Chemistry.

Answer

There are a lot of correct answers here! The point here is that you notice that Inorganic Chemistry is a very broad field. It has something for almost everyone because many other fields overlap with Inorganic Chemistry. You might notice that

some of the sub-fields you identified are also interdisciplinary fields between inorganic chemistry and another discipline. For a list of some of the subfields of Inorganic Chemistry, check this [Wikipedia article](#).

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1.2: Inorganic vs Organic Chemistry

The division between the fields of Inorganic and Organic chemistry has become blurred. For example, let's look at one of the major classes of catalysts used for organic synthesis reactions: organometallic catalysts (Figure 1.2.1). Organometallic catalysts like these, and all organometallic compounds, contain metals that are bonded to carbon or carbon-containing molecules. So, are they "inorganic" because they contain metals, or "organic" because they contain carbon? These illustrate that clear divisions between organic and inorganic chemistry do not exist. Further, metal ions are common in biology and so the idea that metals are "inorganic" and thus classed as "non-living or non-biological" is incorrect. A canonical example is the organometallic catalyst adenosylcobalamin, which is an important biological cofactor containing a cobalt (Co) ion (Figure 1.2.1, right) and a cobalt-carbon bond.

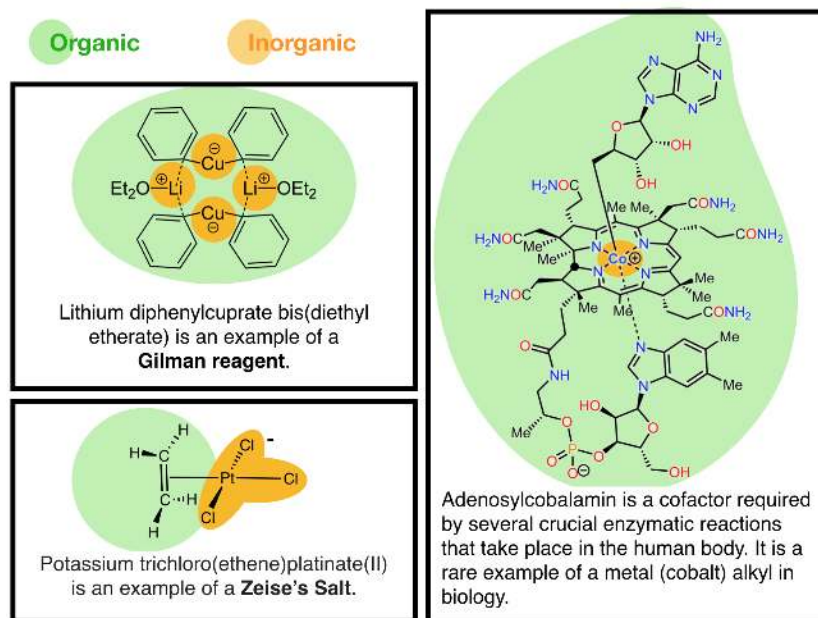


Figure 1.2.1: Some examples of organometallic catalysts. These compounds catalyze organic reactions or biochemical reactions and they are compounds that contain both carbon and metals. These compounds are examples of molecules that cannot be defined only as organic molecules or only as inorganic molecules. Adenosylcobalamin is an example of an organometallic catalyst that is present in biology, further illustrating that "inorganic" metals are important cofactors in biology. This image is based on information on the Wikipedia article on Organometallic Chemistry and is created from images found there; attribution to images created by [Alsosaid1987](#), [AdoCbl-ColorCoded](#), [CC BY-SA 4.0](#) and [Smokefoot](#), [Zeise'sSalt](#), [CC BY-SA 3.0](#). (Kathryn Haas; CC-BY-SA)

Some of the subfields of Inorganic Chemistry focus on electrical conductivity of inorganic materials (i.e., conduction, superconduction, and semiconduction) and on the study of optical and electronic properties of inorganic nanomaterials. Electrical conductivity is a canonical property of metals, but carbon-based materials also demonstrate electrical conductivity. For example, carbon nanotubes conduct electricity through their extended conjugated π systems. Fullerenes, of which the most famous is Buckminsterfullerene, or Buckeyball (C_{60}), demonstrate interesting properties that are similar to nanoparticles, and when combined with metals and crystallized can demonstrate superconductivity.

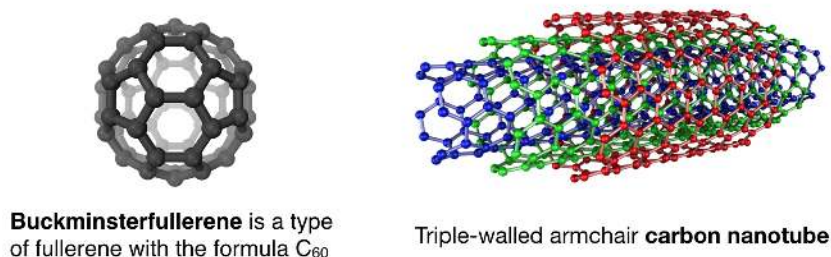


Figure 1.2.2: This figure is created from information found on the Wikipedia articles for [Buckminsterfullerene](#) and [carbon nanotubes](#). Attribution [Eric Wieser](#), [Multi-walled Carbon Nanotube](#), [CC BY-SA 3.0](#). (Kathryn Haas; CC-BY-SA)

Although carbon nanotubes and fullerenes are allotropes of carbon, their material properties are somewhat foreign to many organic chemists, who traditionally have focused on smaller organic molecules having very different properties. However, these properties are familiar to inorganic chemists. Thus, inorganic chemists have embraced these molecules as "inorganic" due to the fact that they behave more like inorganic materials than smaller organic molecules. This class of carbon-based molecules serves as another example of molecules that are not perfectly matched to the traditional definitions of "organic" and "inorganic" chemistry. Certainly, the future will hold more and more examples of molecules that do not fit into the traditional disciplines of chemistry.

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1.3: History of Inorganic Chemistry

Metals serve an essential role in many aspects of human civilization and have defined Ages of human history. The period of time from about 3300 BC to 1200 BC is often referred to as the Bronze Age. During this period our ancestors first started using metal and learned to mix various elements with copper to make a strong alloy, called bronze. This Age yielded significant advancement in crafting of sharper knives and stronger weapons out of metal instead of rock, wood and bone. Around 1200 BC the human race found an even harder metal and discovered a much stronger alloy called steel. This period is known as the Iron Age. More recently, periods of time known as Gold Rushes have caused huge changes in population distributions and wealth in some countries. Metal has obvious importance in our modern way of life. Today, iron and steel are used for making buildings, machines, automobiles, jewelry, cooking pots, tools, weapons, vehicles, electronics, surgical instruments and symbolic structures like the Eiffel Tower. Gold, silver, and copper still serve as currency for trade and exchange of goods and services.

The existence of Chemistry as a field of study owes much to the fact that gold was a valuable commodity throughout our history. In both the ancient Egyptian society and during the Roman Empire, the gold mines were the property of the state, not an individual or group. So there were few ways for most people to legally get any gold for themselves. The Alchemists were a varied group of scholars and charlatans who aimed to solve this problem by creating the **Philosopher's Stone** (which caused the transmutation of lead into gold). Three major streams of alchemy are known, Chinese, Indian, and European, with all three streams having some factors in common. Techniques developed in the European stream ultimately influenced the development of the science of chemistry.

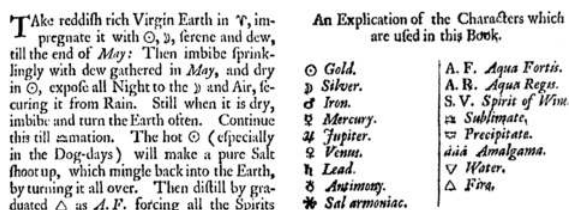


Figure 1.3.1. Alchemist recipe. Many of the specific approaches that alchemists used when they tried changing lead into gold are vague and unclear. Each alchemist had his own code for recording his data. The processes were kept secret so others could not profit from them. Different scholars developed their own set of symbols as they recorded the information they came up with. Many alchemists were not very honest, taking money from a nobleman by claiming to be able to make gold from lead, then leaving town in the middle of the night. Sometimes the nobleman would detect the fraud and have the alchemist hung. By the 1300's, several European rulers had declared alchemy to be illegal and set out strict punishments for those practicing the alchemical arts.

Although alchemists were never successful in changing lead into gold, they made several contributions to modern-day chemistry. Strong acids and bases were discovered, including nitric acid (H_2NO_3), sulfuric acid (H_3SO_4), and hydrochloric acid (HCl), and sodium hydroxide (NaOH). Glassware for running chemical reactions were developed, as well as methods for distillation, crystallization, sublimation. Alchemy helped improve the study of metallurgy and the extraction of metals from ores. More systematic approaches to research were developed that allowed the discovery of atoms and laid the groundwork for development of the periodic table. For more about the History of Chemistry in general, try the LibreText page [History of Chemistry](#).

Inorganic compounds have been known and used since antiquity; probably the oldest is the deep blue pigment called Prussian blue, $\text{Fe}_4[\text{Fe}(\text{CN})_6]$. However, the chemical nature of these substances was unknown until the late nineteenth and early twentieth century when the modern field of Coordination Chemistry emerged. Much of what we know about inorganic chemistry is based largely on the work of and debates between of Alfred Werner (1866–1919; Nobel Prize in Chemistry in 1913) and Sophus Mads Jørgensen (1837–1914). After Werner succeeded in these debates, the field of Inorganic Chemistry declined in popularity until the mid twentieth century when the second world war stimulated renewed interest. During the post-war era, several important discoveries and theories were developed. For example, important theories of bonding in coordination compounds were developed. Soon after World War II, Crystal Field Theory (CFT) and Ligand Field Theory (LFT) were developed. These are two critical and complimentary theories that provide explanation of spectroscopic, chemical, and structural properties of inorganic coordination compounds; CFT being more simple, and LFT more accurate. In the 1950's, organometallic catalysts were discovered that catalyzed important organic reactions.



The **Haber-Bosch** Process is catalyzed by an inorganic oxide catalyst and is one of the world's most important industrial reactions. It provides for the synthesis of ammonia directly from elemental nitrogen, N_2 , and hydrogen, H_2 . Since its development in the early

twentieth century, it has led to the production of an enormous quantity of fertilizer, vastly increasing global food production. As a result, it is estimated that a significant fraction of the nitrogen content in the typical human body is ultimately derived from this process. Yet while the reaction must be run at high temperatures and pressures in the industrial setting, the nitrogenase enzyme on the roots of plants can carry out this reaction at the mild conditions within soil. Intense investigations were then aimed to improve inorganic catalysts through understanding the metal cofactors in enzymes. The link between the Haber-Bosch industrial process and the nitrogenase enzyme was an early bridge between the fields of organometallic chemistry and biochemistry.

Contributions

Part of this page is adapted from K.Haas Ph.D. dissertation

Part of this page is adapted from the following LibreTexts pages

- [Alchemy](#) contributed by CK-12 Foundation by Sharon Bewick, Richard Parsons, Therese Forsythe, Shonna Robinson, and Jean Dupon.
- [Werner's Theory of Coordination Compounds](#)
- [The Haber-Bosch Process Worksheet](#)

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CHAPTER OVERVIEW

2: Atomic Theory

This chapter reviews atomic orbitals and the periodic properties of atoms.

Learning Objectives

- Understand the sizes, shapes, and relative energies of atomic orbitals and how they depend on quantum numbers
- Identify the number of radial and angular nodes in an atomic orbital
- Determine the ground state electron configuration and number of unpaired electrons for a given atom
- Explain periodic trends and their physical origins

Thumbnail illustrates the electron density of a 3d orbital, generated with the [Orbitron](#)

[2.1: Quantum Numbers and Atomic Wavefunctions](#)

[2.2: Shielding](#)

[2.3: Aufbau Principle](#)

[2.4: Periodic Properties of Atoms](#)

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2.1: Quantum Numbers and Atomic Wavefunctions

Considering the failures of the Bohr model in systems with more than one electron, Erwin Schrödinger and Werner Heisenberg proposed a major change in the paradigm regarding the electron. In several breakthrough papers (1925-1927) they attributed wave properties to electrons and they each received Nobel prizes for developing the theories of Wave Mechanics (or the "New Quantum Mechanics"). This approach treated electrons as having "dual" nature: possessing properties of *both* waves and particles.

Schrödinger's Equation describes the behavior of the electron (in a hydrogen atom) in three dimensions. It is a mathematical equation that defines the electron's position, mass, total energy, and potential energy. The simplest form of the Schrödinger Equation is as follows:

$$\hat{H}\psi = E\psi \quad (2.1.1)$$

where $\hat{H}$ is the Hamiltonian operator, E is the energy of the electron, and ψ is the wavefunction.

The Hamiltonian, $\hat{H}$

The Hamiltonian operator is like a set of instructions that tells us what to do with the function that follows it. A Hamiltonian operator is a function over three-dimensional space that corresponds to the sum of kinetic energies and potential energies of the particles in a system, one electron and its nucleus in this case. The Hamiltonian operator for a one-electron system is:

$$\hat{H} = \frac{-h^2}{8\pi^2m_e} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) - \frac{Ze^2}{4\pi\epsilon_0 r} \quad (2.1.2)$$

Where h is Planck's constant, m_e is the mass of the electron, e is the charge of the electron, r is the distance from the nucleus ($r = \sqrt{x^2 + y^2 + z^2}$), Z is the charge of the nucleus, and $4\pi\epsilon_0$ is permittivity of a vacuum.

Kinetic Energy

The first part of the Hamiltonian written above, $\frac{-h^2}{8\pi^2m_e} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)$ describes the kinetic energy of the electron. This is the energy due to motion of the electron.

Potential energy

The second part written above, $\frac{-Ze^2}{4\pi\epsilon_0 r}$ describes the potential energy of the electron, and is commonly written as $V(r)$ or $V(x, y, z)$.

$$V(x, y, z) = \frac{-Ze^2}{4\pi\epsilon_0 r} = \frac{-Ze^2}{4\pi\epsilon_0 \sqrt{x^2 + y^2 + z^2}} \quad (2.1.3)$$

The potential energy depends on the attractive electrostatic force between the electron and the nucleus. You might notice that this attraction is essentially the same as the electrostatic force defined by Coulomb's law. And, just as in Coulomb's law, when two opposite charges are attracted to one another, the potential energy of the force is negative. Thus, when an electron is close to the nucleus, the potential energy is a large negative number corresponding to a strong attractive force. When an electron is farther from the nucleus, the potential energy is still negative but with a smaller magnitude, corresponding to a weaker attractive force. If the electron is very far from the nucleus ($r = \infty$) then the attractive force, and the potential energy, is zero.

The Wavefunction, ψ

In simple terms, the wavefunction (ψ) of an electron describes the electron's position in space, relative to the nucleus. The square of the ψ describes an atomic orbital. We can't define the position too exactly because we would violate the Heisenberg Uncertainty principle, but we can define its wave. Here, we will describe the ψ in general terms. Generally, in a one-electron atom, the electron ψ is defined by the wave's distance from the nucleus and its angle with respect to x, y, and z axes of the atom's Cartesian coordinates (nucleus is at the origin). The general form of the (ψ) for an electron in a hydrogen atom can be written as follows:

$$\psi_{n,l,m_l} = R_{n,l}(r) + Y_{l,m_l}(\theta, \phi) \quad (2.1.4)$$

Quantum numbers

In one dimension, the wavefunction requires only one quantum number, n . A full explanation of the three-dimensional wavefunction for an electron is outside of the scope of this course. Instead, we will focus on gaining a conceptual understanding by extending knowledge of the one-dimensional wavefunction to a three-dimensional space. Extending the wavefunction to three dimensions requires a total of three quantum numbers. In addition to n , which we saw in the one-dimensional case, we also need l and m_l to express the wavefunction in three dimensions. A complete solution to the Schrödinger equation, both the three-dimensional wavefunction and energy, includes a set of three quantum numbers (n, l, m_l) . The wavefunction describes what we know as an atomic orbital; it defines the region in space where the electron is located. Additionally, there is a fourth quantum number, m_s . The m_s quantum number accounts for the observed interaction of electrons with an applied magnetic field; it is an additional postulate that is not part of the wavefunction. These four quantum numbers are described below.

The quantum number, n : This is the principle quantum number. This number represents the shell, including both the overall energy of the electron in that shell and the size of that shell. An allowed value for n is any non-zero, positive integer (1, 2, 3, 4, ... etc are allowed, but 4.1 is not allowed).

The quantum number, l : This is the angular momentum quantum number that corresponds to the subshell and its shape. It represents the angular dependence of the subshell, or the "shape" of the orbitals within a subshell. The allowed values of l depends on n . The allowed values of l for an electron in shell n are integer values between 0 to $(n - 1)$, or $l = 0 \rightarrow (n - 1)$. These values correspond to the orbital shape where $l = 0$ is an s-orbital, $l = 1$ is a p-orbital, $l = 2$ is a d-orbital, $l = 3$ is an f-orbital.

The quantum number m_l : This is the magnetic quantum number. Its possible values give the the number of orbitals within a subshell and its specific value gives the orbital's orientation in space. The allowed values of m_l depend on the value of l . The value of m_l is allowed to be any positive or negative integer between $+l$ and $-l$. In other terms, $m_l = (0, \pm 1 \dots \pm l)$. For example, if the electron is in a 3-p-orbital, then $n = 3, l = 1$, and the possible values of m_l are $-1, 0$, and $+1$. Since there are three possible values of m_l there are three orbitals in the p subshell. The specific m_l value defines in which of the three possible p-orbitals (p_x, p_y , or p_z) the electron exists. In the case of the s subshell, there is only one value, $m_l = 0$ because $l = 0$. The one value corresponds to the fact that there is only one s orbital in any shell.

The quantum number m_s : This quantum number accounts for the electron's "spin". In short, electrons interact with magnetic fields in a way that is similar to how a tiny bar magnet would interact with a magnetic field. The allowed values for m_s are $+\frac{1}{2}$ and $-\frac{1}{2}$.

Table 2.1.1. Quantum Numbers

SYMBOL	NAME	VALUES	MEANING
n	principle	1, 2, 3...(any integer)	energy level, orbital size
l	angular momentum	$0 \rightarrow (n - 1)$	subshell, 0 = s, 1 = p, 2 = d, 3 = f... this is the angular dependence of the orbital, shape of the orbital *letters have historical meaning, sharp, principle, diffuse, fundamental
m_l	magnetic	$0, \pm 1 \dots \pm l$	orientation of angular momentum in space, orbital orientation
m_s	spin	$+\frac{1}{2}, -\frac{1}{2}$	orientation of the electron's magnetic field "spin up" or "spin down"

Atomic orbitals (one electron wavefunctions)

One simplified representation of the three-dimensional wavefunction is shown below. This representation breaks the wavefunction into two parts: the **radial contribution** ($R_{n,l}(r)$) and the **angular contribution** ($Y_{l,m_l}(\theta, \phi)$).

$$\psi_{(n,l,m_l)} = R_{n,l}(r) \times Y_{l,m_l}(\theta, \phi) \quad (2.1.5)$$

Radial contribution, $R_{n,l}(r)$

The radial part of the wavefunction, $R_{n,l}(r)$ gives the radial variation of ψ . In other words, $R_{n,l}(r)$ defines how the wavefunction depends on the distance of the electron from the nucleus (the radius). The principle quantum number n is derived from the radial part of the wavefunction, and determines the size (radial extent) of an orbital. The $R_{n,l}(r)$ parts of the wavefunction for a hydrogenic atom are plotted in Figure 2.1.1. Notice that the $R_{n,l}(r)$ of all s-orbitals (solid lines) reaches a maximum at $r = 0$. This is unique to the s-orbitals' $R_{n,l}(r)$. The $R_{n,l}(r)$ of p- and d-orbitals approaches zero as r approaches zero. This has important consequences for how closely an electron in these orbitals can approach the nucleus.

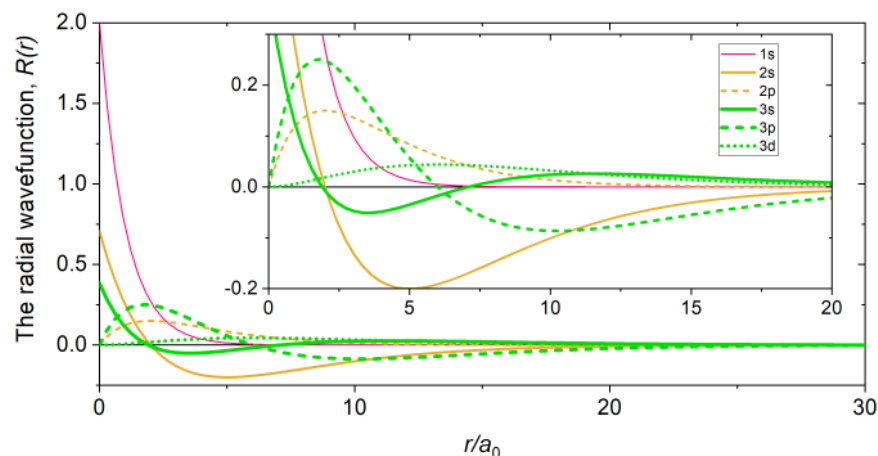


Figure 2.1.1: Plots of the radial wavefunction, $R_{n,l}(r)$, for the first three shells. The wavefunctions are plotted relative to $\frac{r}{a_0}$, where $a_0 = 52.9 \text{ pm} = 0.529 \text{ \AA}$ is the Bohr Radius (the radius of a hydrogen 1s orbital). The expressions for these radial wavefunctions ($R_{n,l}(r)$) are shown in Table 2.1.2.

Radial probability function, $4\pi r^2 (R_{n,l}(r))^2$

The square of the wavefunction is proportional to the probability of finding a particle (electron) at some point in space. The square of the radial part of the wavefunction is called the radial distribution function $4\pi r^2 (R_{n,l}(r))^2$, and it describes the probability of locating the electron at some distance r away from the nucleus. When we normalize the probability functions by dividing the function by its integral over all space, we get the plots shown in Figure 2.1.2. The normalized probability functions are compared to the original radial part of the wavefunctions in Figure 2.1.3. The most probable distance for finding an electron is shown by the maximum value of the function. For an electron in the 1s orbital of H, the most probable distance from the nucleus occurs at $r = 1a_0$. This is the Bohr Radius, and it has a value of ($a_0 = 52.9 \text{ pm} = 0.529 \text{ \AA}$). It is convenient to plot the functions of the hydrogen atomic orbitals relative to the size of its smallest orbital, the 1s orbital; this is the reason we plot $R_{n,l}(r)$ and $4\pi r^2 (R_{n,l}(r))^2$ relative to $\frac{r}{a_0}$.

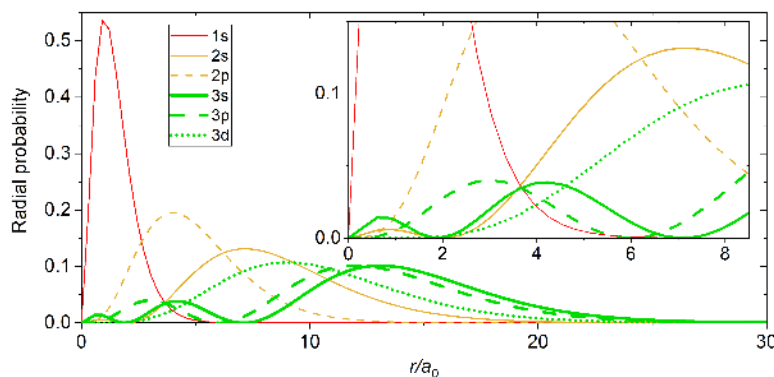


Figure 2.1.2: Plots of the normalized radial probability functions for the first three shells.

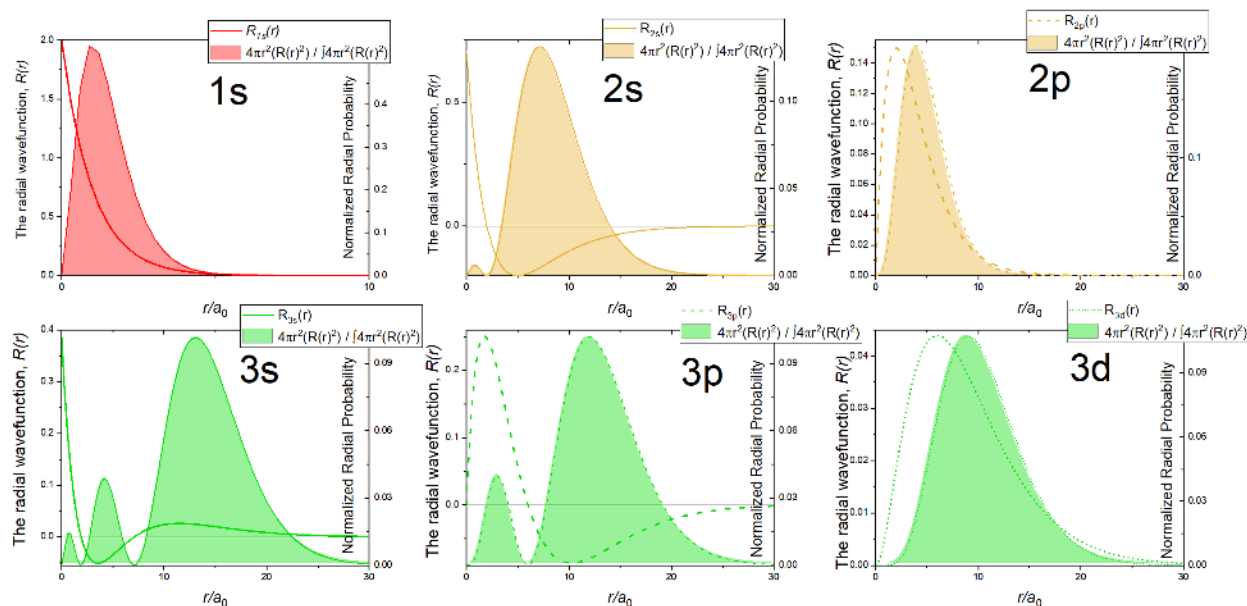


Figure 2.1.3. Plots of the normalized radial probability function for each orbital in the first three shells (shaded curves). The y-axis on the left of each plot corresponds to the radial wavefunction (line); the y-axis on the right of each plot corresponds to radial probability function (shaded).

Radial nodes

In the case of the 3-dimensional wavefunction, there are two different types of nodes: radial nodes and angular nodes. **Radial nodes occur where the radial part of the wavefunction is zero ($R(r) = 0$).** These are easy to see by plotting the radial part of the wavefunction and finding where the radial part of the wavefunction (and the radial probability function) is zero as seen in Figures 2.1.2 and 2.1.3. These nodes are spherical in shape and depend on the energy level and subshell (the values of n and l). **The number of radial nodes is $n - l - 1$.** An example of a radial node is the single node that occurs in the $2s$ orbital ($2 - 0 - 1 = 1$ node). In contrast, the $1s$ orbital has zero radial nodes ($1 - 0 - 1 = 0$ nodes). Where there is a node, there is zero probability of finding an electron.

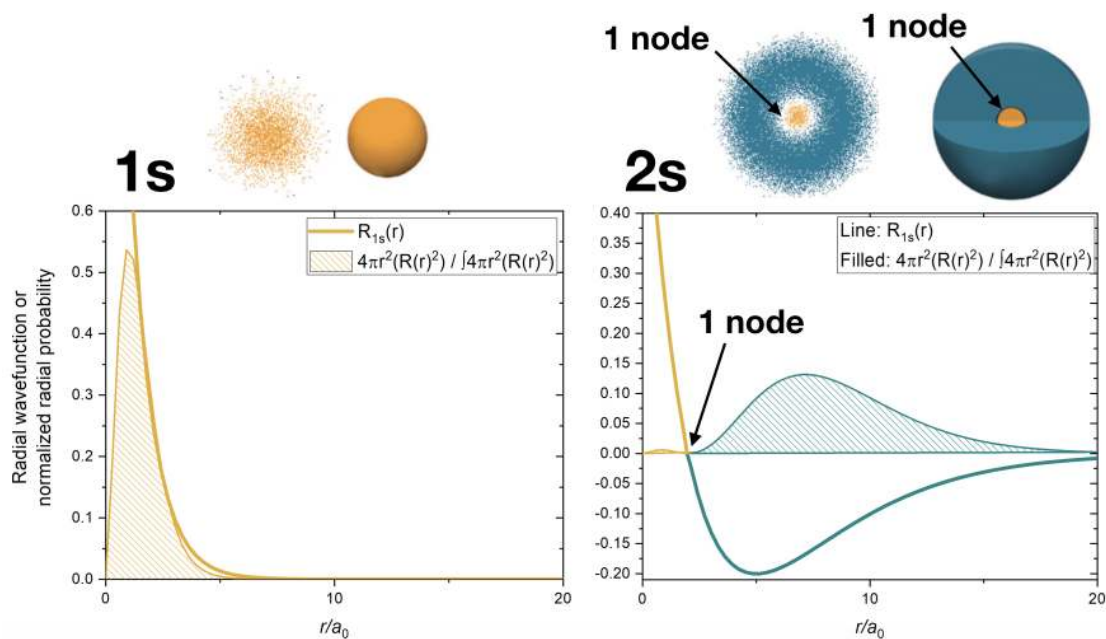


Figure 2.1.4: Visualization of the 1s and 2s atomic orbitals. Each orbital is shown as both an electron probability density plot and a contour plot above its wavefunction and probability density function. The 1s orbital has zero radial nodes. The 2s orbital has one radial node where its wavefunction changes sign and its radial probability function is zero. Modified by K. Haas (CC-BY-NC-SA; Libretexts)

Boundary surfaces

A true node occurs where the probability of finding an electron is zero. Nodes occur where $\psi = \psi^2 = 0$. Far from the nucleus, the probability of finding the electron rapidly approaches zero, but is never *exactly* zero. What this means is that it is rather improbable to find the electron at far distances from the nucleus, but its not impossible. The rapid fall of probability creates a boundary surface rather than a node. The boundary surface represents the area around the nucleus where the electron exists most of the time.

Exercise 2.1.1

How many radial nodes are in the s, p, d, and f orbitals in the first four shells? $n = 1, 2, 3, 4$?

Answer

You can approach this answer by using the mathematical relationship between the number of radial nodes and the values of n and l : **number of radial nodes is $n - 1 - l$** . You could also notice the pattern that the first orbital of any type (1s, 2p, 3d...) has zero radial nodes, the second orbital of a type (2s, 3p, 4d...) has one radial node, the third orbital of a type has two radial nodes (3s, 4p, 5d...)...etc.

A full list of all of the orbitals in the first four shells and their number of radial nodes is below. Its a good idea to prove to yourself that the numbers given below are consistent with the plots of the orbitals wavefunctions for those orbitals in the first three shells:

s-orbitals:

1s: $n - 1 - l = 1 - 1 - 0 = 0$, zero radial nodes

2s: $n - 1 - l = 2 - 1 - 0 = 1$, one radial node

3s: $n - 1 - l = 3 - 1 - 0 = 2$, two radial nodes

4s: $n - 1 - l = 4 - 1 - 0 = 3$, three radial nodes

p-orbitals

2p: $n - 1 - l = 2 - 1 - 1 = 0$, zero radial nodes

3p: $n - 1 - l = 3 - 1 - 1 = 1$, one radial node

4p: $n - 1 - l = 4 - 1 - 1 = 2$, two radial nodes

d-orbitals

3d: $n - 1 - l = 3 - 1 - 2 = 0$, zero radial nodes

4d: $n - 1 - l = 4 - 1 - 2 = 1$, one radial nodes

f-orbital

4f: $n - 1 - l = 4 - 1 - 3 = 0$, zero radial nodes

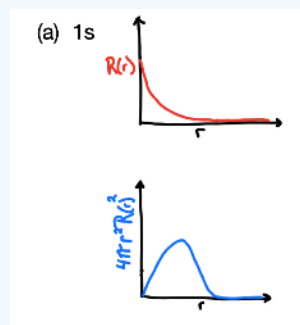
Exercise 2.1.2

Draw a rough plot of the following:

- The radial wavefunction, $R_{n,l}(r)$ and the probability density function, $4\pi r^2(R_{n,l}(r))^2$, for the 1s orbital.
- The radial wavefunction, $R_{n,l}(r)$ and the probability density function, $4\pi r^2(R_{n,l}(r))^2$, for the 4s orbital.

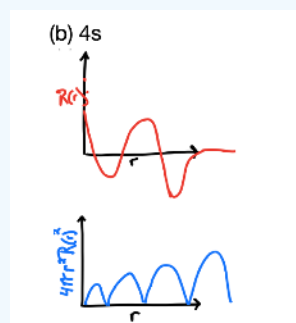
Answer, 1s

The 1s orbital's $R_{n,l}(r)$ reaches a maximum near the origin and approaches zero far from the nucleus. The $4\pi r^2(R_{n,l}(r))^2$ approaches zero at the origin, has a maximum intensity at the Bohr radius, and approaches zero far from the nucleus (its boundary surface).



Answer, 4s

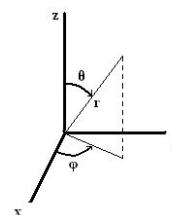
Like all s orbitals, the 4s orbital's $R_{n,l}(r)$ reaches a maximum near $X=0$, but it would be less intense than the function of the 1s, 2s, and 3s orbitals. It should have three radial nodes and so its wavefunction would change sign three times. This would give four regions of electron density in the probability function while the function would approach zero (boundary surface) far from the nucleus.



Angular contribution, $Y_{l,m_l}(\theta, \phi)$, and angular probability function, $(Y_{l,m_l}(\theta, \phi))^2$

The angular contribution to the wavefunction, $Y_{l,m_l}(\theta, \phi)$, describes the wavefunction's shape, or the angle with respect to a coordinate system. To describe the direction in space, we use spherical coordinates that tell us distance *and* orientation in 3-dimensional space. There are three spherical coordinates: r , ϕ , and θ . r is the radius, or the actual distance from the origin. ϕ and θ are angles. ϕ is measured from the positive x axis in the xy plane and may be between 0 and 2π . θ is measured from the positive z

axis towards the xy plane and may be between 0 and π . $Y_{l,m_l}(\theta, \phi)$, is slightly more difficult to describe than the radial contribution was. This is partly because $Y_{l,m_l}(\theta, \phi)$ contains imaginary numbers, which have no real, physical meaning. However, the angular part of the wavefunction becomes more "real" when you square it to get angular probability density, a more tangible concept described as the **shapes of orbitals**.



The values of θ , ϕ , and $y(\theta, \phi)$ for orbitals in the hydrogen atom are listed in Table 2.1.2, but $Y_{l,m_l}(\theta, \phi)$ is only a mathematical function and has not real physical meaning. The square of the radial wavefunction, $Y_{l,m_l}(\theta, \phi)^2$ gives the probability of finding the electron at a point in space on a ray described by (ϕ, θ) . Thus $Y_{l,m_l}(\theta, \phi)^2$ describes the shape of the orbital.

Table 2.1.2. Components of the angular wavefunction and the resulting orbital shapes. The plots in the two right hand columns were created using Desmos and Mathematica: specifically "Spherical Harmonics" from the Wolfram Demonstrations Project, <http://demonstrations.wolfram.com/SphericalHarmonics/> and "Visualizing Atomic Orbitals", <https://demonstrations.wolfram.com/VisualizingAtomicOrbitals/>.

subshell	m_l (orbital)	θ	ϕ	Plots of θ^2	orbital shapes
$l = 0$, s	$m_l = 0$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$		
$l = 1$, p	$m_l = 0$	$\frac{\sqrt{6}}{2} \cos \theta$	$\frac{1}{\sqrt{2\pi}}$		
$l = 1$, p	$m_l = +1$	$\frac{\sqrt{3}}{2} \sin \theta$	$\frac{1}{\sqrt{2\pi}} e^{i\phi}$		
$l = 1$, p	$m_l = -1$	$\frac{\sqrt{3}}{2} \sin \theta$	$\frac{1}{\sqrt{2\pi}} e^{-i\phi}$		
$l = 2$, d	0	$\frac{\sqrt{10}}{4} (3 \cos^2 \theta - 1)$	$\frac{1}{\sqrt{2\pi}}$		
$l = 2$, d	$m_l = +1$	$\frac{\sqrt{15}}{2} \sin \theta \cos \theta$	$\frac{1}{\sqrt{2\pi}} e^{i\phi}$		
$l = 2$, d	$m_l = -1$	$\frac{\sqrt{15}}{2} \sin \theta \cos \theta$	$\frac{1}{\sqrt{2\pi}} e^{-i\phi}$		

subshell	m_l (orbital)	θ	ϕ	Plots of θ^2	orbital shapes
$l = 2, d$	$m_l = +2$	$\frac{\sqrt{15}}{4} \sin^2 \theta$	$\frac{1}{\sqrt{2\pi}} e^{i2\phi}$		
$l = 2, d$	$m_l = -2$	$\frac{\sqrt{15}}{4} \sin^2 \theta$	$\frac{1}{\sqrt{2\pi}} e^{-i2\phi}$		

Angular nodes

Angular nodes exist where $(Y_{l,m_l}(\theta, \phi))^2 = 0$. These nodes are planar in shape, and they depend on the value of l . The number of angular nodes in any orbital is equal to l . This means that s-orbitals ($l = 0$) have zero angular nodes, p-orbitals ($l = 1$) have one angular node, d-orbitals ($l = 2$) have two angular nodes, and so on. Planar nodes can be flat planes (like the nodes in all p orbitals) or they can have a conical shape, like the two angular nodes in the d_{z^2} orbital. Angular nodes in some p and d orbitals are shown in Figure 2.1.5.

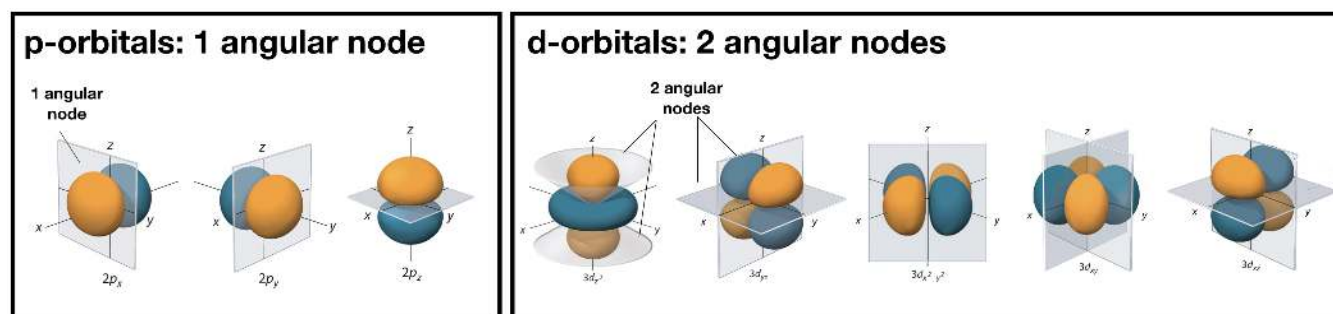


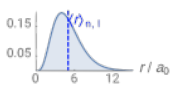
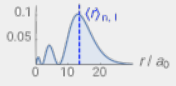
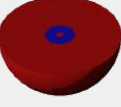
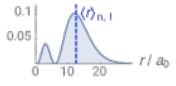
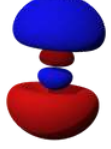
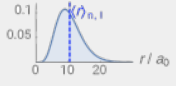
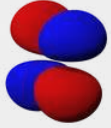
Figure 2.1.5: The angular nodes in p-orbitals and d-orbitals are shown. Every p orbital has one angular node, and every d orbital has two angular nodes. The number of angular nodes in any orbital is equal to the value of the quantum number, l . (CC BY-NC-SA; anonymous by request)

Atomic Orbitals

Atomic orbitals result from a combination of both the radial and angular contributions of the wavefunction. Atomic orbitals can have both angular nodes and radial nodes, depending on the values of n and l .

The chart below compares the radial variation, angular variation, and their combinations (orbitals).

Orbital	Radial Probability	Radial Nodes	Angular Probability	Angular Nodes	Combination (Orbital)	Total Nodes
1s		0		0		0
2s		1		0		1

Orbital	Radial Probability	Radial Nodes	Angular Probability	Angular Nodes	Combination (Orbital)	Total Nodes
2p		0		1		1
3s		2		0		2
3p		1		1		2
3d		0		2		2

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2.2: Shielding

Introduction

Coulomb's Law is from classical physics; it tells us that particles with opposite electrostatic charge are attracted to each other; and the larger the charge on either particle or the closer the distance between them, the stronger the attraction. Coulomb's law explains why atomic size decreases as the charge on the nucleus increases, but it can't explain the nuances and variations in size as we go across the periodic table. Coulomb's Law also explains why electrons in different shells (n), at different distances from the nucleus, have different energies. But on its own, Coulomb's law doesn't quite explain why electron subshells within a shell (like 2s vs 2p) would have different energies. To explain these things, we need to consider how both electron **shielding** and **penetration** result in variations in effective nuclear charge (Z^*) that depend on shell and subshell.

Effective Nuclear Charge (Z^*)

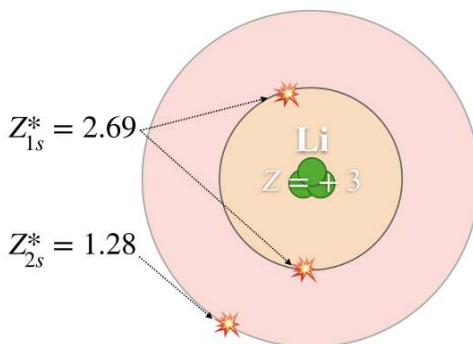


Figure 2.2.1. In a lithium atom, the nuclear charge (Z) is +3. 1s electrons experience an effective nuclear charge (Z^*) of +2.69, and 2s electrons experience an Z^* of +1.28. (CC-BY-NC-SA; Kathryn Haas)

Coulombs' law works well for predicting the energy of an electron in a hydrogen atom (because H has only one electron). It also works for hydrogen-like atoms: any nucleus with exactly one electron (a He^+ ion, for example, has one electron). However, Coulomb's law is insufficient for predicting the energies of electrons in multi-electron atoms and ions.

Electrons within a multi-electron atom interact with the nucleus and with all other electrons. Each electron in a multi-electron atom experiences both attraction to the nucleus and repulsion from interactions with other electrons. The presence of multiple electrons decreases the nuclear attraction to some extent. Each electron in a multi-electron atom experiences a different magnitude of (and attraction to) the nuclear charge depending on what specific shell and subshell the electron occupies. The amount of positive nuclear charge experienced by any individual electron is the **effective nuclear charge (Z^*)**.

For example, in lithium (Li), none of the three electrons "feel" the full +3 charge from the nucleus (see Figure 2.2.1). Rather, each electron "feels" a Z^* that is less than the actual Z and that depends on the electron's orbital. The actual nuclear charge in Li is $Z = +3$; the 1s electrons experience a $Z^* = +2.69$, and the 2s electron experiences a $Z^* = +1.28$. In general, core electrons (or the electrons closest to the nucleus), "feel" a Z^* that is close to, but less than, Z . On the other hand, outer valence electrons experience a Z^* that is much less than Z .

In summary:

- Core electrons: $Z^* \approx Z$
- Valence electrons: $Z^* \ll Z$

Shielding:

Shielding is the reduction of true nuclear charge (Z) to the effective nuclear charge (Z^*) by other electrons in a multi-electron atom or ion. Shielding occurs in all atoms and ions that have more than one electron. H is the only atom in which shielding does not occur.

Explanation of shielding: Electrons in a multi-electron atom interact with the nucleus and all other electrons in the atom. To describe shielding, we can use a simplified model of the atom: we will choose an electron-of-interest in a multi-electron atom and treat all the "other" electrons as a group of spherically-distributed negative charge. Classical electrostatics allows us to treat spherical distribution of charge as a point of charge at the center of the distribution. Thus, we consider all the "other" electrons in our atom as a point of negative charge in the center of the atom. While the positive charge of the nucleus provides an attractive force toward our electron, the negative charge distribution at the center of the atom would provide a repulsive force. The attractive and repulsive forces would partially cancel each other; but since there are less "other" electrons than there are protons in our atom, the nuclear charge is never completely canceled. The "other" electrons partially block, or shield, part of the nuclear charge so that our electron-of-interest experiences a partially-reduced nuclear charge, the Z^* .

In reality, there is not a one-for-one "canceling" of the nuclear charge by each electron. Partly due to *penetration*, no single electron can completely shield a full unit of positive charge. Core electrons shield valence electrons, but valence electrons have little effect on the Z^* of core electrons. The ability to shield, and be shielded by, other electrons strongly depends the electron orbital's average distance from the nucleus and its *penetration*; thus shielding depends on both shell (n) and subshell (l).

Shielding depends on electron penetration

Coulomb's law shows us that distance of an electron from its nucleus is important in determining the electron's energy (its attraction to the nucleus). The shell number (n) determines approximately how far an electron is from the nucleus *on average*. Thus, all orbitals in the same shell (s,p,d) have similar sizes and similar average distance of their electrons from the nucleus. But there is another distance-related factor that plays a critical role in determining orbital energy levels: *penetration*. **Penetration** describes the ability of an electron in a given subshell to penetrate into other shells and subshells to get close to the nucleus. Penetration is the **extent to which an electron can approach the nucleus**. Penetration depends on both the shell (n) and subshell (l).

The penetration of individual orbitals can be visualized using the radial probability functions. For example, Figure 2.2.2 below shows plots of the radial probability function of the 1s, 2s, and 2p orbitals. From these plots, we can see that the 1s orbital is able to approach the closest to the nucleus; thus it is the most penetrating. While the 2s and 2p have most of their probability at a farther distance from the nucleus (compared to 1s), the 2s orbital and the 2p orbital have different extents of penetration. Notice that the 2s orbital is able to penetrate the 1s orbital because of the central 2s lobe. The 2p orbital penetrates somewhat into the 1s, but it cannot approach the nucleus as closely as the 2s orbital can. While the 2s orbital penetrates more than 2p (2s orbital can approach closer to the nucleus), the 2p is slightly *closer on average* than 2s. The order of Z^* in 2s and 2p subshells depends on which factor (average distance or penetration) is more important. In the first two rows of the periodic table, penetration is the dominant factor that results in 2s having a lower energy compared to 2p (see Figure 2.2.4 for values).

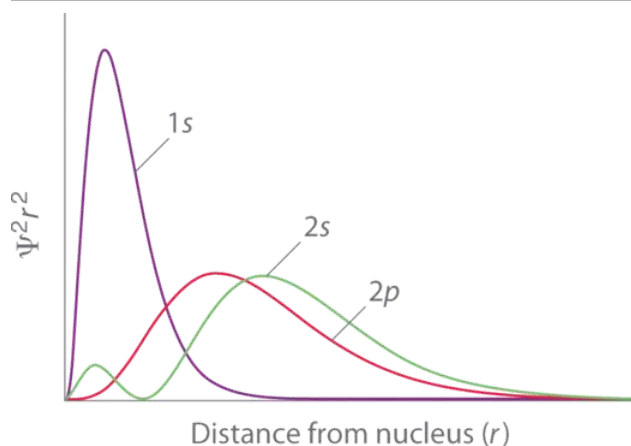


Figure 2.2.2. Orbital Penetration. A comparison of the radial probability distribution of the 2s and 2p orbitals for various states of the hydrogen atom shows that the 2s orbital penetrates inside the 1s orbital more than the 2p orbital does. Consequently, when an electron is in the small inner lobe of the 2s orbital, it experiences a relatively large value of Z^* , which causes the energy of the 2s orbital to be lower than the energy of the 2p orbital.

An electron orbital's penetration effects its ability to shield other electrons and effects the extent to which it is shielded by other electrons. In general, electron orbitals that have greater penetration experience stronger attraction to the nucleus and less shielding

by other electrons; these electrons thus experience a larger (Z^*). Electrons in orbitals that have greater penetration also shield other electrons to a greater extent.

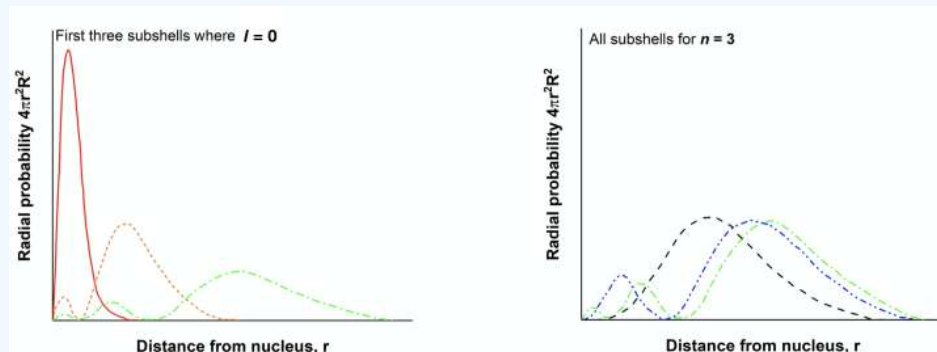
Within the same shell value (n), the penetrating power of an electron follows this trend in subshells (l):

$$s > p > d > f$$

Exercises

Exercises

- Compare the 2s and 2p orbitals:
 - Which is closer to the nucleus on average?
 - Which is more penetrating?
 - Which orbital experiences a stronger Z^* and is thus lower in energy (consider your experience, but also inspect Figure 2.2.5)? Please explain.
- Visit the [Orbitron](#) and peruse the radial distribution functions for different orbitals. Compare the 2p and 3s orbitals:
 - Which is farther from the nucleus on average?
 - Which is more penetrating?
 - Which orbital is lower in energy?
- Which atom, Li, or N, has a stronger valence Z^* ? Explain why.
- Explain why 2s and 2p subshells are completely degenerate in a hydrogen atom.
- Which atom has a smaller radii: Be or F? Explain.
- Which electrons shield others more effectively: 3p or 3d?
- Use the clues given in the figures below to label the radial distribution functions shown.



- Explain why the ground state (most energetically favorable) electron configuration of Be is $1s^2 2s^2$ rather than alternative configurations like $1s^2 2s^1 2p^1$ or $1s^2 2p^2$.

Answer 1

- The 2s orbital is closer to the nucleus on average.
- The 2s orbital is more penetrating than 2p.
- You might "know" that the 2s orbital is lower in energy than 2p because 2s fills first. But a close inspection of [Figures 1.1.2.3 and 1.1.2.4 from the previous page](#) indicates that while the 2s and 2p elements are degenerate in Ne (element 10), for elements with atomic number 11 and greater 2p has a higher Z^* than 2s! This example illustrates that both average distance and penetration are factors in determining Z^* , and the factor that is more important may change as we increase in atomic number.

Answer 2

- The 3s orbital reaches farther away from the nucleus and is on average farther from the nucleus than 2p.
- The 3s orbital is more penetrating than 2p, even though 3s is farther on average!
- The 2p orbital is lower in energy than 3s; this is because 2p is still significantly closer to the nucleus on average and experiences a stronger Z^* . (Penetration is not the only consideration!)

Answer 3

A nitrogen atom has a stronger effective nuclear charge (Z^*) than lithium due to its greater number of protons; even though N also has more electrons that would shield the nuclear charge, each electron only partially shields each proton. This means that atoms with greater atomic number always have greater Z^* for any given electron.

Answer 4

The hydrogen atom has only one electron; thus there is no shielding to consider. When there are not other electrons to shield the nucleus, penetration and shielding are irrelevant, and subshells within a shell are degenerate.

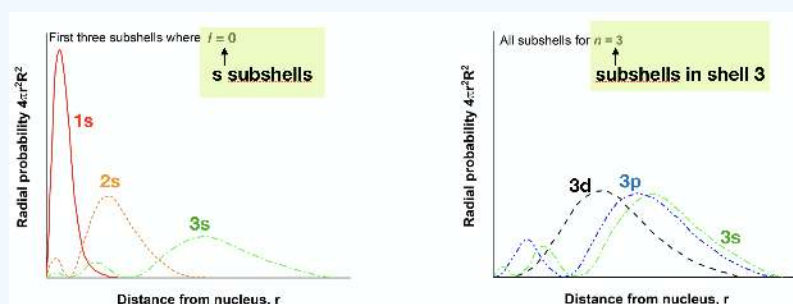
Answer 5

Fluorine has a smaller radii than Beryllium due to F having a greater valence Z^* and therefore pulling the valence electrons close to the nucleus and providing a smaller atomic radii.

Answer 6

3p shields better than 3d because p orbitals penetrate more than d orbitals within the same shell.

Answer 7

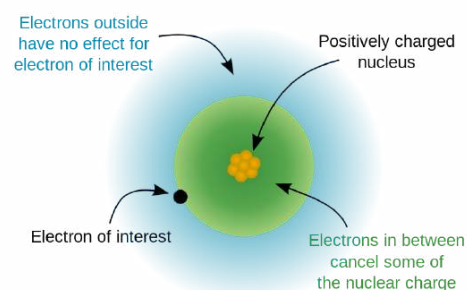


Answer 8

This question is asking why the 2s orbital fills in Be before 2p is occupied. This is a multi-electron atom, therefore core electrons shield the 2s and 2p orbitals to different extents. In Be we expect the 2s orbital to fill before 2p because 2s penetrates more and experiences a higher Z^* .

Slater's rules for estimating Z^*

Figure 2.2.3. Diagram illustrating effective nuclear charge according to Slater's rules.

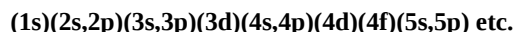


The Z^* can be estimated using a number of different methods; probably the best known and most commonly used method is known as **Slater's Rules**. Slater developed a set of rules to estimate Z^* based on how many other electrons exist in the atom and on the orbital location of the electron-of-interest. These two factors are important determinants in shielding, and they are used to calculate a **shielding constant (S)** used in Slater's formula:

$$Z^* = Z - S \quad (2.2.1)$$

where Z is the actual nuclear charge (the atomic number) and Z^* is the effective nuclear charge.

In the calculation of S , it is assumed that electrons closer to the nucleus than our electron-of-interest cancel some of the nuclear charge; those farther from the nucleus have no effect. To calculate S , all the relevant orbitals in an atom are written out in order of increasing energy, separating them into "groups". Each change in shell number is a new group; s and p subshells are in the same group but d and f orbitals are their own group. You write out all the orbitals using parentheses until you get to the group of the electron-of-interest, like this:



****Critical: The orbitals must be written in order of increasing energy!**

1. Electrons in the same Group():

- Each other electron (not counting the electron-of-interest) in the same group () as the chosen electron, contributes 0.35 to S .
- Each 1s electron contributes 0.30 to S .
Conceptually, this means electrons in the same group shield each other 30-35%.

2. Electrons in Groups() to the left:

- If the electron-of-interest is in a d or f subshell, every electron in groups () to the left contributes 1.00 to S .
Conceptually, this means that d and f electrons are shielded 100% by all electrons in the same shell with a smaller value of l , as well as all electrons in lower shells (n).
- If the electron-of-interest is in an s or p subshell, all electrons in the next lower shell ($n - 1$) contribute 0.85 to S . And all the electrons in even lower shells contribute 1.00 to S .
Conceptually, this means that s and p electrons are shielded 85% by the electrons one shell lower, and 100% by all electrons in shells $n - 2$ or lower.

3. Electrons in Groups() to the right:

- Electrons to the right contribute 0 to S .
Conceptually, this means electrons in higher energy groups don't contribute to shielding.

[A video explaining how to use Slater's Rules](#)

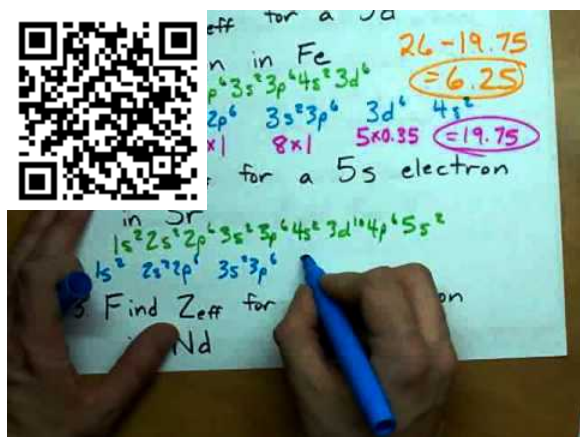
Using Slater's Rules: 3 Examples
chemistNATE

1s² 2s²2p⁶ 3s²3p⁶ 3d⁴ 4s²
 2x1 8x1 8x1 5x0.35 = 6.25
 = 19.75

2 Find Z_{eff} for a 5s electron
 in Sr
 1s²2s²2p⁶3s²3p⁶4s²3d¹⁰4p⁶5s²
 1s² 2s²2p⁶ 3s²3p⁶

3 Find Z_{eff} for
 Nd

Watch on YouTube



Example 2.2.1: Fluorine, Neon, and Sodium

What is the Z^* experienced by the valence electrons in the three isoelectronic species: fluorine anion (F^-), neutral neon atom (Ne), and sodium cation (Na^+)?

Solution

Each species has 10 electrons, and the number of core electrons is 2 (10 total electrons - 8 valence), but the effective nuclear charge varies because each has a different atomic number (Z). The approximate Z^* can be found with Slater's Rules. For all of these species, we would calculate the same sigma value:

Calculating S : $(1s)(2s,2p)$, $S = 2(0.85) + 7(0.35) = 1.7 + 2.45 = 4.15$

Fluorine anion: $Z^* = 9 - S = 9 - 4.15 = 4.85$

Neon atom: $Z^* = 10 - S = 10 - 4.15 = 5.85$

Sodium Cation: $Z^* = 11 - S = 11 - 4.15 = 6.85$

So, the sodium cation has the greatest effective nuclear charge.

Exercise 2.2.1

Calculate Z^* for a 3d-electron in a zinc (Zn) atom.

Answer

Write out the relevant orbitals: $(1s)(2s,2p)(3s,3p)(3d) (4s)$

Notice that although 4s is fully occupied, we don't include it because in Zn, 4s is higher in energy than 3d. Therefore, 4s is to the right of the d electrons we are considering. The electron-of-interest is in 3d, so the other nine electrons in 3d each contribute 0.35 to the value of S . The other 18 electrons each contribute 1 to the value of S .

$$S = 18(1) + 9(0.35) = 21.15$$

$$Z^* = 30 - 21.15 = 8.85$$

So, although the nuclear charge of Zn is 30, the 3d electrons only experience a $Z^* \approx 8.85$!

"Best" values for Z^*

Slater's rules are a set of simple rules for predicting S and Z^* based on empirical evidence from quantum mechanical calculations. In other words, the Z^* calculated from Slater's rules are approximate values. The values considered to be the most accurate are derived from quantum mechanical calculations directly. You can find these values in a nice chart on the Wikipedia article of Effective Nuclear Charge. The chart is recreated here in Figure 2.2.4 for convenience:

Z or Z _{eff}	H																	He
Z	1.00																	2.00
1s	1.00																	1.69
	Li	Be											B	C	N	O	F	Ne
Z	3.00	4.00											5.00	6.00	7.00	8.00	9.00	10.00
1s	2.69	3.69											4.68	5.67	6.67	7.66	8.65	9.64
2s	1.28	1.91											2.58	3.22	3.85	4.49	5.13	5.76
2p													2.42	3.14	3.83	4.45	5.10	5.76
	Na	Mg											Al	Si	P	S	Cl	Ar
Z	11.00	12.00											13.00	14.00	15.00	16.00	17.00	18.00
1s	10.63	11.61											12.59	13.58	14.56	15.54	16.52	17.51
2s	6.57	7.39											8.21	9.02	9.83	10.63	11.43	12.23
2p	6.80	7.83											8.96	9.95	10.96	11.98	12.99	14.01
3s	2.51	3.31											4.12	4.90	5.64	6.37	7.07	7.76
3p													4.07	4.29	4.89	5.48	6.12	6.76
	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Z	19.00	20.00	21.00	22.00	23.00	24.00	25.00	26.00	27.00	28.00	29.00	30.00	31.00	32.00	33.00	34.00	35.00	36.00
1s	18.49	19.47	20.46	21.44	22.43	23.41	24.40	25.38	26.37	27.35	28.34	29.33	30.31	31.29	32.28	33.26	34.25	35.23
2s	13.01	13.78	14.57	15.38	16.18	16.98	17.79	18.60	19.41	20.21	21.02	21.83	22.60	23.37	24.13	24.89	25.64	26.40
2p	15.03	16.04	17.06	18.07	19.07	20.08	21.08	22.09	23.09	24.10	25.10	26.10	27.09	28.08	29.07	30.07	31.06	32.05
3s	8.68	9.60	10.34	11.03	11.71	12.37	13.02	13.68	14.32	14.96	15.59	16.22	17.00	17.79	18.60	19.40	20.22	21.03
3p	7.73	8.66	9.41	10.10	10.79	11.47	12.11	12.78	13.44	14.09	14.73	15.37	16.20	17.01	17.85	18.71	19.57	20.43
4s	3.50	4.40	4.63	4.82	4.98	5.13	5.28	5.43	5.58	5.71	5.84	5.97	7.07	8.04	8.94	9.76	10.55	11.32
3d			7.12	8.14	8.98	9.76	10.53	11.18	11.86	12.53	13.20	13.88	15.09	16.25	17.38	18.48	19.56	20.63
4p													6.22	6.78	7.45	8.29	9.03	9.34
	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Z	37.00	38.00	39.00	40.00	41.00	42.00	43.00	44.00	45.00	46.00	47.00	48.00	49.00	50.00	51.00	52.00	53.00	54.00
1s	36.21	37.19	38.18	39.16	40.14	41.13	42.11	43.09	44.08	45.06	46.04	47.03	48.01	48.99	49.97	50.96	51.94	52.92
2s	27.16	27.90	28.62	29.37	30.13	30.88	31.63	32.38	33.16	33.88	34.63	35.39	36.12	36.86	37.60	38.33	39.07	39.80
2p	33.04	34.03	35.00	35.99	36.98	37.97	38.94	39.95	40.94	41.93	42.92	43.91	44.90	45.89	46.87	47.86	48.85	49.84
3s	21.84	22.66	23.55	24.36	25.17	25.98	26.79	27.60	28.44	29.22	30.03	30.84	31.63	32.42	33.21	34.00	34.79	35.58
3p	21.30	22.17	23.09	23.85	24.62	25.47	26.38	27.22	28.15	29.02	29.81	30.69	31.52	32.35	33.18	34.01	34.84	35.67
4s	12.39	13.44	14.26	14.90	15.28	16.10	17.20	17.66	18.58	18.99	19.87	20.87	21.76	22.66	23.54	24.41	25.30	26.17
3d	21.68	22.73	25.40	25.57	26.25	27.23	28.35	29.36	30.41	31.45	32.54	33.61	34.68	35.74	36.80	37.84	38.90	39.95
4p	10.88	11.93	12.75	13.46	14.08	14.98	15.81	16.44	17.14	17.72	18.56	19.41	20.37	21.27	22.18	23.12	24.03	24.96
5s	4.99	6.07	6.26	6.45	5.92	6.11	7.23	6.49	6.64	(empty)	6.76	8.19	9.51	10.63	11.62	12.54	13.40	14.22
4d			15.96	13.07	11.24	11.39	12.88	12.81	13.44	13.62	14.76	15.88	16.94	17.97	18.97	19.96	20.93	21.89
5p													8.47	9.10	10.00	10.81	11.61	12.43

Figure 2.2.4. This chart shows Z^* values calculated by Clementi et. al. in 1963 and 1967 that are consistent with SCF Calculations. This chart was created using data from the Wikipedia article on Effective Nuclear Charge.

Z^* modulates attraction

When valence electrons experience less nuclear charge than core electrons, different electrons experience different magnitudes of attraction to the nucleus. A modified form of Coulomb's Law is written below, where e is the charge of an electron, Z^* is the effective nuclear charge experienced by that electron, and r is the radius (distance of the electron from the nucleus).

$$F_{eff} = k \frac{Z^* e^2}{r^2} \quad (2.2.2)$$

This formula suggests that if we can estimate Z^* , then we can predict the attractive force experienced by, and the energy of, an electron in a multi-electron atom (ex. Li).

The attraction of the nucleus to valence electrons determines the atomic or ionic size, ionization energy, electron affinity, and electronegativity. The stronger the attraction, and the stronger Z^* , the closer the electrons are pulled toward the nucleus. This in turn results in a smaller size, higher ionization energy, higher electron affinity, and stronger electronegativity.

General Periodic Trends in Z^*

Close inspection of Figure 2.2.4 and analysis of Slater's rules indicate that there are some predictable trends in Z^* . The data from Figure 2.2.4 is plotted below in Figure 2.2.5 to provide a visual aid to the discussion below.

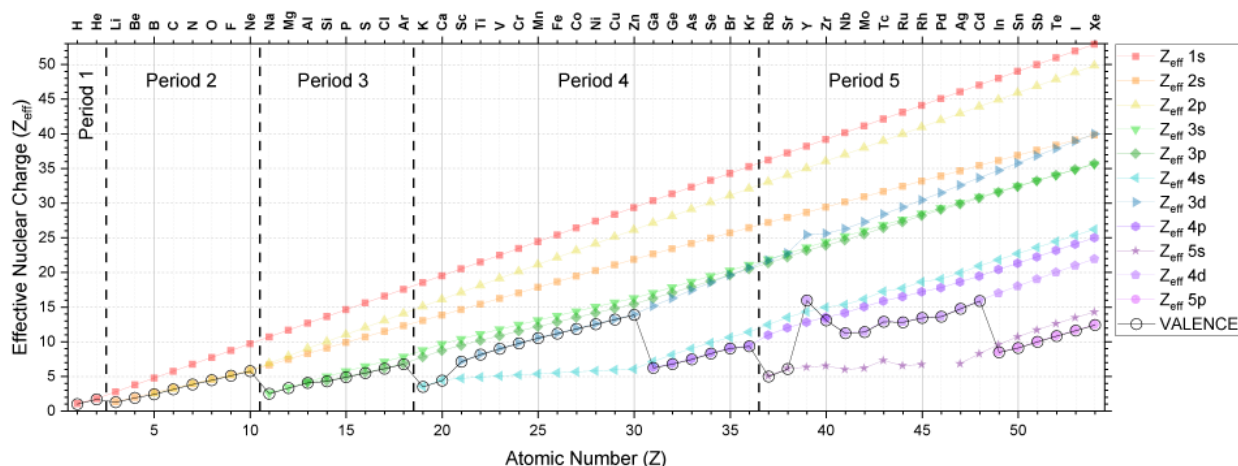


Figure 2.2.5: The Z^* values for electrons in each subshell of the first 54 elements (H to Xe) are plotted. Each subshell is plotted as a different series (see legend) and the valence shell is highlighted by a solid black line with open circles. (CC-BY-NC-SA; Kathryn Haas)

Trends in Z^* for electrons in a specific shell and subshell

The Z^* for electrons in a given shell and subshell generally increases as atomic number increases; this trend holds true going across the periodic table and down the periodic table. Convince yourself that this is true for any subshell by examining Figure 2.2.5.

Do you notice any exceptions to this general trend?

Inspection of Figure 2.2.5 should confirm for you that the Z^* increases as Z increases for electrons in any subshell (like the 1s subshell for example, which is plotted above as a red line with square points). You can see this trend as the positive slope in each series. There is one obvious exception in Period 5 in elements 39 (Y) to 41 (Nb); the Z^* of 4s actually decreases across these three elements as atomic number increases. There is also an exception between Y and Zr in the 3d subshell, and between Tc and Ru in the 5s subshell.

For valence electrons:

It is useful to understand trends in valence Z^* because the valence Z^* determines atomic/ionic properties and chemical reactivity. The trends in the valence Z^* are not simple because as atomic number increases, the valence shell and/or subshell also changes. The valence Z^* is indicated in Figure 2.2.5 as a black line with open circles.

Down the table: As we go down a column of the periodic table, the valence Z^* increases. This is a simple trend because type of subshell is consistent and there is an increase only in shell and in atomic number, Z . This trend is best illustrated by inspection of Figure 2.2.4.

Across the table: the trend depends on shell and subshell, but generally Z^* increases across a period.

Periods 1-3 (s and p only): As we go across the table in periods 1-3, the shell stays constant as Z increases and the subshell changes from s to p. In these periods, there is a gradual increase in valence Z^* as we move across any of the first three periods.

Periods 4 and 5 (s, p, and d): Now we have some more complex trends because valence subshell and shell are changing as we increase in atomic number. Notice that the valence Z^* generally increases going across a period as long as subshell isn't changing; the exception is within the 4d subshell (elements 39-44 or Y-Ru). In general, going from $(n)s$ subshell to $(n-1)d$ subshell, there is a relatively large increase in valence Z^* . And in going from $(n-1)d$ subshell to $(n)p$ subshell, there is a relatively large decrease in Z^* .

From one period to another: From Figure 2.2.5, we can see that as we increase Z by one proton, going from one period to the next, there is a relatively large decrease in Z^* (from Ne to Na, for example). This is because as Z increases by a small interval, the shell number increases, and so the electrons in the valence shell are much farther from the nucleus and are more shielded by all the electrons in the lower shell numbers.

Exercises

Exercise 2.2.2

1. Compare trends in Z^* and atomic size. Explain how and why atomic size depends on Z^* .
2. Compare trends in Z^* and ionization energy. Explain how and why ionization energy depends on Z^* .

Answer

1. On the periodic table, atomic radius generally decreases across the periods (left to right) and increases down the groups. As atomic number increases across the periodic table, nuclear charge (Z) increases and Z^* increases. In turn, the atomic radius decreases because the higher nuclear charge (and thus higher Z^*) pulls electrons closer to the nucleus. Atomic radius increases down the periodic table because the shell number increases. Despite an increase in Z^* going down the periodic table, larger atomic radii results from electrons occupying higher shells.
2. Ionization energies (IE) are inversely related to atomic radius; IE increases across the periods and decreases down the groups. Since the nucleus holds valence electrons more strongly (due to higher Z^*) across the periods, IE increases because valence electrons are harder to remove. Down the periodic table, larger atomic radius causes electrons in valence orbitals to be shielded by core electrons. Recall that shielding reduces the nuclear charge available to electrons in higher orbital levels, resulting in a lower Z^* . With more shielding and lower Z^* , the valence electrons are held less tightly by the nucleus such that ionization energy decreases (i.e. valence electrons are easier to remove).

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2.3: Aufbau Principle

Introduction

The **Aufbau Principle** (also called the **building-up principle** or the **Aufbau rule**) states that, in the ground state of an atom or ion, electrons fill atomic orbitals of the lowest available energy level before occupying higher-energy levels. In general, an electron will occupy an atomic orbital with the lowest value of n, l, m_l , in that order of priority. The value of m_s for an unpaired electron is conventionally assigned a value of $+\frac{1}{2}$. Each electron in an atom/ion must have a unique set of values for all four quantum numbers.

The Ground State of Hydrogenic Atoms/Ions

In a hydrogenic atom/ion, there is only one electron. In this case, the *only factor determining energy is the value of n* . The ground state will always be the 1s orbital ($n = 1, l = 0, m_l = 0, m_s = +\frac{1}{2}$).

The Ground State of Multi-electron Atoms/Ions

In atoms/ions with two or more electrons, the ground state electron configuration (1) minimizes the total energy of the electrons, (2) obeys the Pauli exclusion principle, (3) obeys Hund's rule of maximum multiplicity, and (4) considers the exchange interaction. These "rules" are described below.

(1) Electrons will occupy the lowest energy orbitals in order to minimize the total energy. The two quantum numbers that are related to energy in multi-electron atoms are n , and l . Thus, orbitals with the lowest values of n and l will fill first.

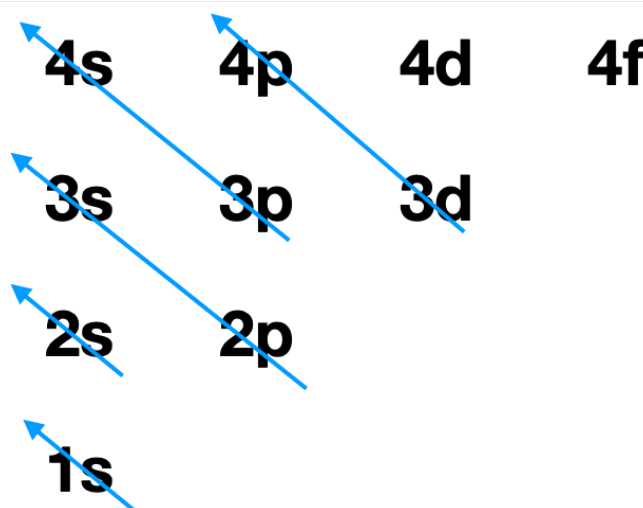
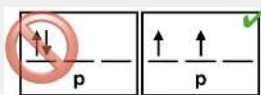


Figure 2.3.1. The order of filling of ground state electron orbitals is 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, etc... (CC-BY-NC-SA; Kathryn Haas)

(2) **Hund's rule of maximum multiplicity** states that for a given electron configuration, the lowest energy arrangement of electrons in degenerate orbitals is the one with the greatest "multiplicity," where multiplicity is the number of unpaired electrons (n) plus 1 (multiplicity = $n + 1$). This rule is used to predict the **ground state** of an atom or molecule with one or more open electronic shells.

Hund's rule is based on empirical observation of atomic spectra, and it is a consequence of the energy required to pair two electrons in the same orbital. This energy of repulsion between two electrons in the same orbitals is a Coulombic energy of repulsion, Π_e , caused by two electrons with like charge sharing the same area of space (an orbital). When more than one electron occupies a set of degenerate orbitals, the most favorable arrangement is one where the number of paired electrons is minimized.

A simplified definition of Hund's rule is that **the lowest energy arrangement is the one with the greatest number of unpaired electrons**. This implies that if two or more orbitals of equal energy are available, electrons will occupy them singly before filling them in pairs. Examples of ground state arrangements of electrons in three degenerate p-orbitals is given in the figure shown here.



# of electrons in a p-subshell	Arrangement (unpaired electrons, n)	Spin Multiplicity ($n + 1$)
1	$\uparrow \quad _ \quad _$ ($n = 1$)	2
2	$\uparrow \quad \uparrow \quad _$ ($n = 2$)	3
3	$\uparrow \quad \uparrow \quad \uparrow$ ($n = 3$)	4
4	$\uparrow\downarrow \quad \uparrow \quad \uparrow$ ($n = 2$)	3
5	$\uparrow\downarrow \quad \uparrow\downarrow \quad \uparrow$ ($n = 1$)	2
6	$\uparrow\downarrow \quad \uparrow\downarrow \quad \uparrow\downarrow$ ($n = 0$)	1

Figure 2.3.2.

Hund's rule is that spin multiplicity must be maximized in the ground state. (CC-BY-NC-SA; Libretexts)

(3) **The Pauli exclusion principle** states that it is impossible for two electrons of a multi-electron atom to have the same set of values for the four quantum numbers. Two electrons in different orbitals will have a different set of n , l , and m_l values. When two electrons reside in the same orbital, they possess the same n , l , and m_l values, therefore their m_s must be different. Thus, two electrons in the same orbital must have opposite half-integer spin projections of $+\frac{1}{2}$ and $-\frac{1}{2}$.

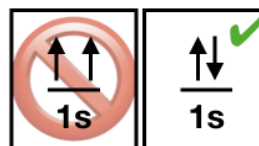


Figure 2.3.3. Pauli's principle tells us that paired electrons must have opposite spin. CC-BY-NC-SA; Libretexts)

(4) The **exchange interaction** is sometimes called the *exchange energy* or *exchange force*, however it is not a true energy or force. Rather, it is a quantum mechanical effect that takes place between identical particles. The exchange interaction results in a ground state electron configuration with unpaired electrons all being of the same spin. Unpaired electrons are conventionally written in the "spin up" direction.

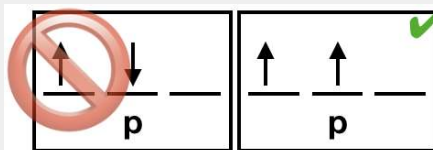


Figure 2.3.4.

Unpaired electrons should have identical spins according due to the exchange interaction. (CC-BY-NC-SA; Libretexts)

Trends in Expected Electron Configuration

The four "rules" above can be used as guidelines to predict the ground state electron configuration of atoms, the filling of subshells, and the configuration of electrons in degenerate orbitals. However, the utility of these guidelines for predicting actual electron configurations requires more nuanced knowledge of the relative energy levels of orbitals.

Generally, orbital energy levels directly correspond to their shell number. Additionally, orbitals within a shell generally follow the energetic trend where $s < p < d < f$. Although these general trends for relative orbital energy levels hold true for most of the main block elements (the s - and p -blocks), there are important exceptions in the orbital energy levels of transition metal atoms and ions of the d - and f -blocks.

Figure 2.3.5. This is a depiction of the periodic table that highlights the s -, p -, d -, and f - blocks in different colors. Violations of the expected trend in electron configurations are outlined in a heavy black line. Several elements of the d block and f block violate the general trends in electron configuration because their orbital energy levels do not follow *general* trends. (CC-BY-NC-SA; Kathryn Haas)

Elements that violate general trends in electron configuration are outlined with a dark line in Figure 2.3.5. All of the exceptions are within the d - and f - blocks, and the violations are caused by an unexpected order of the orbital energy levels.

In the next section you will learn why the orbital energy levels correlate with shell number and why subshells within a shell usually follow the trend that $s < p < d < f$. You will also learn why there are occasional exceptions to this trend and how these exceptions

influence elemental properties.

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2.4: Periodic Properties of Atoms

Introduction

General periodic trends are specific patterns that are present within the periodic table; these are patterns in properties like electronegativity, ionization energy, electron affinity, atomic radius, melting point, and metallic character. General periodic trends provide chemists with an invaluable tool to quickly predict an element's properties. These trends exist because of the similar atomic structure of the elements within their respective group families or periods, and because of the periodic nature of the elements. Some of the general periodic trends are described in this section.

Ionization Energy

Ionization energy (IE) is the energy required to **remove an electron from a neutral atom or cation in its gaseous phase**. IE is also known as *ionization potential*.



Conceptually, ionization energy is the affinity of an element for its outermost electron (an electron it already has in its valence shell).

1st, 2nd, and 3rd Ionization Energies

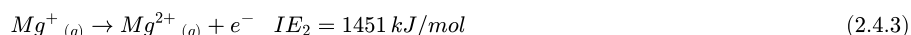
The symbol IE_1 stands for the **first ionization energy** (energy required to take away an electron from a neutral atom, where $n = 0$). The symbol IE_2 stands for the **second ionization energy** (energy required to take away an electron from an atom with a +1 charge, $n = 2$.)

First Ionization Energy, IE_1 (general element, A): $A_{(g)} \rightarrow A_{(g)}^{1+} + e^{-}$

Second Ionization Energy, IE_2 (general element, A): $A_{(g)}^{1+} \rightarrow A_{(g)}^{2+} + e^{-}$

Third Ionization Energy, IE_3 (general element, A): $A_{(g)}^{2+} \rightarrow A_{(g)}^{3+} + e^{-}$

Each succeeding ionization energy is larger than the preceding energy. This means that $IE_1 < IE_2 < IE_3 < \dots < IE_n$ will always be true. For example, ionization energy increases as succeeding electrons are taken away from Mg.



Ionization energy is correlated with the strength of attraction between the positively-charged nucleus and the negatively-charged valence electrons, and therefore the effective nuclear charge Z_{eff} . The higher the ionization energy, the stronger the attractive force between nucleus and valence electrons, and the more energy is required to remove a valence electron. The lower the ionization energy, the weaker the attractive force between nucleus and valence electrons, and the less energy required to remove a valence electron.

General periodic trends in ionization energy

In general, ionization energies increase from left to right and decrease down a group; however there are variations in these trends that would be expected from the effects of penetration and shielding. The trends in **first ionization energy** are shown in Figure 2.4.1 and are summarized below.

- **Across a period:** As Z^* increases across a period, the ionization energy of the elements generally increases from left to right. However there are breaks or variation in the trends in the following cases:
 - IE is especially low when removal of an electron creates a newly empty p subshell (examples include IE_1 of B, Al, Sc)
 - IE energy is especially low where removal of an electron results in a half-filled p or d subshell (examples include IE_1 of O, S)
 - IE increases more gradually across the d- and f-subshells compared to s- and p- subshells. This is because d- and f- electrons are weakly penetrating and experience especially low Z^* .

- **From one period to the next:** There is an especially large decrease in IE with the start of every new period (from He to Li or from Ne to Na for example). This is consistent with the idea that IE is especially low when removal of an electron creates a newly empty s-subshell.
- **Noble gases:** The noble gases possess very high ionization energies. Note that helium has the highest ionization energy of all the elements.
- **Down a group:** Although Z^* increases going down a group, there is no reliable trend in IE going down any group; in some cases IE increases going down a group, while in other cases IE decreases going down a group.

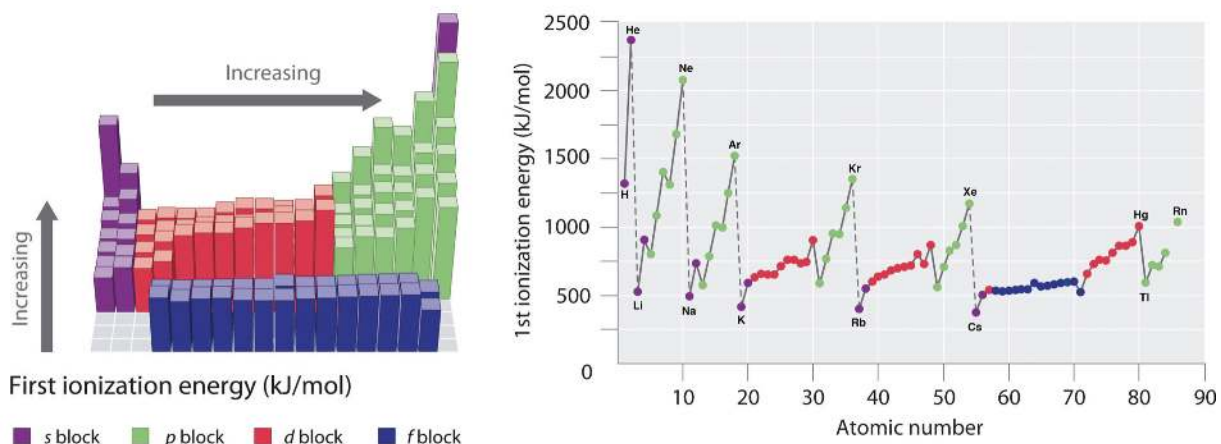


Figure 2.4.1: Graph showing the First Ionization Energy of the Elements. Left: ionization energies are mapped onto the periodic table, where the magnitude of ionization energy is depicted as a bar graph. Right: the first ionization energy is plotted against atomic number. In both panels the elements of the s-block are shown in purple, p-block are shown in green, d-block are shown in red, and f-block are shown in blue.

Plots of the IE_1 , IE_2 , and IE_3 of elements from hydrogen to krypton (first four periods) are shown in Figure 2.4.2. Notice that $IE_3 > IE_2 > IE_1$. Also notice that trends mentioned above for IE_1 hold true for subsequent ionizations when electron configurations are considered!

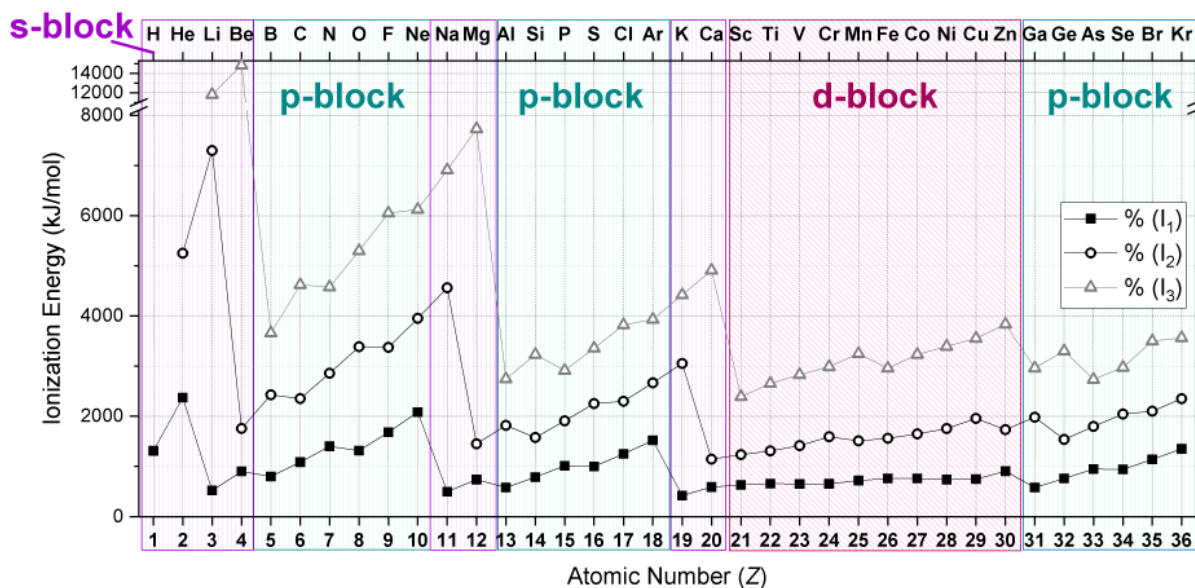


Figure 2.4.2. The first (I_1), second (I_2), and third (I_3) ionization energies are plotted for elements with $Z = 1$ to 36 (H to Kr). The position of each element in its atomic form is indicated as s- p- or d-block. (CC-BY-NC-SA; Kathryn Haas)

Electron Affinity

According to IUPAC, there are two different, but equivalent, definitions of **electron affinity (EA)**.⁶

EA defined as removal of an electron

Electron affinity can be defined as the energy *required* when an electron is removed from a gaseous anion. The reaction as shown in equation 2.4.4 is *endothermic* (positive ΔU) for elements except noble gases and alkaline earth metals. Under this definition, the more positive the EA value, the higher an atom's affinity for electrons.



The reaction shown in equation 2.4.4 is similar those that define ionization energy. For this reason, the EA is also described as the *zeroth ionization energy*.

EA defined as addition of an electron

An alternate and more common definition is the microscopic reverse of equation 2.4.4. This more common definition states that electron affinity is the energy *released* when an electron is added to a gaseous atom, as shown in equation 2.4.5. The reaction as shown in equation 2.4.5 is *exothermic* (negative ΔU) for elements except noble gases and alkaline earth metals. The more negative this EA value, the higher an atom's affinity for electrons.



Conceptually, this second definition is quite similar to the concept of electronegativity; but unlike electronegativity, EA is a well-defined quantitative measurement.

Trends in Electron Affinity

For this discussion, we will use the first definition of EA: a more positive (larger) value means that the EA is higher (meaning stronger affinity toward an electron).

Across a period: Similar to ionization energy, EA generally increases across a row of the periodic table; this observation is consistent with the increase in effective nuclear charge (Z^*) from left to right across a period. However, there are variations across a period that are similar to variations in ionization energy and that can be explained by shielding, penetration, and electron configuration.

Down a group: Like the case of ionization energy trends, EA does not consistently decrease going down a column of the periodic table despite the fact that Z^* increases down a group.

The trend in EA follows a similar zig-zag pattern as the one seen with ionization energies, except that it is displaced by one unit from the trend in IE_1 , two units from IE_2 , and so on. For example, EA peaks at F, while IE_1 peaks at Ne, IE_2 peaks at Na, and IE_3 peaks at Mg. A plot of EA for the first 13 elements is shown overlaid on plots of IE_1 , IE_2 and IE_3 in Figure 2.4.3., where the shifts in the peaks and valleys within each zig-zag trend are indicated.

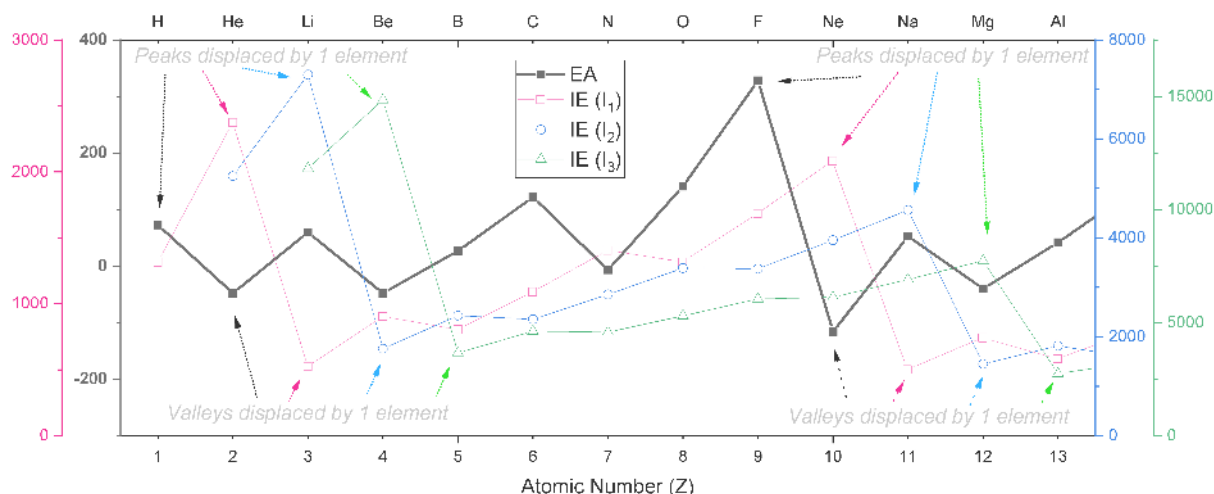


Figure 2.4.3. A plot of electron affinity (the zeroth ionization energy) is overlaid on plots of the first (I_1), second (I_2), and third (I_3) ionization energies. Each plot is shown using separate y-axes. Trends in EA are similar to those in ionization energies, except the peaks and valleys of the trends are shifted by one unit, as indicated. These plots are shown in units of kJ/mole. (CC-BY-NC-SA; Kathryn Haas)

Atomic and Ionic Radii

There are several methods that can be used to determine radii of atoms and ions:

- **Nonpolar atomic radii:** The radius of an atom is derived from the bond lengths within nonpolar molecules; one-half the distance between the nuclei of two atoms within a covalent bond.
- **van der Waals radius:** The radius of an atom is determined by collision with other atoms.
- **Crystal Radii:** The atomic or ionic radius is determined using electron density maps from X-ray data.

The measurement of atomic or ionic size will depend on a number of factors, including the covalent character of bonding in any particular molecule, coordination number, physical state (liquid, solid, gas), the identity of nearby atoms/ions, variation in crystal structure, and distortions within regular crystal structures. You should keep in mind that the size of an atom or ion is a "fuzzy" measure, and the radius under a different set of conditions will probably change slightly.

Regardless, measured atomic and ionic radii reveal obvious trends across the periodic table and between atoms and ions. The relative atomic sizes shown in Figure 2.4.4 were derived from crystallographic data.⁹

Trends in Atomic Radius

Atomic size generally decreases gradually from left to right across a period of elements. As nuclear charge (Z) increases, we expect the effective nuclear charge (Z^*) of the valence electrons to also increase. Increasing Z pulls electrons closer to the nucleus. However, with each additional unit of Z , there is also an additional electron. The change in size is a balance of a compression caused by increasing Z and an expansion in the number of electrons. As a result, the atomic radius decreases *gradually* across a period.

Atomic size generally increases going down a group. As valence electrons occupying higher level shells due to the increasing quantum number (n), size increases despite the fact that Z and Z^* are increasing going down the group.

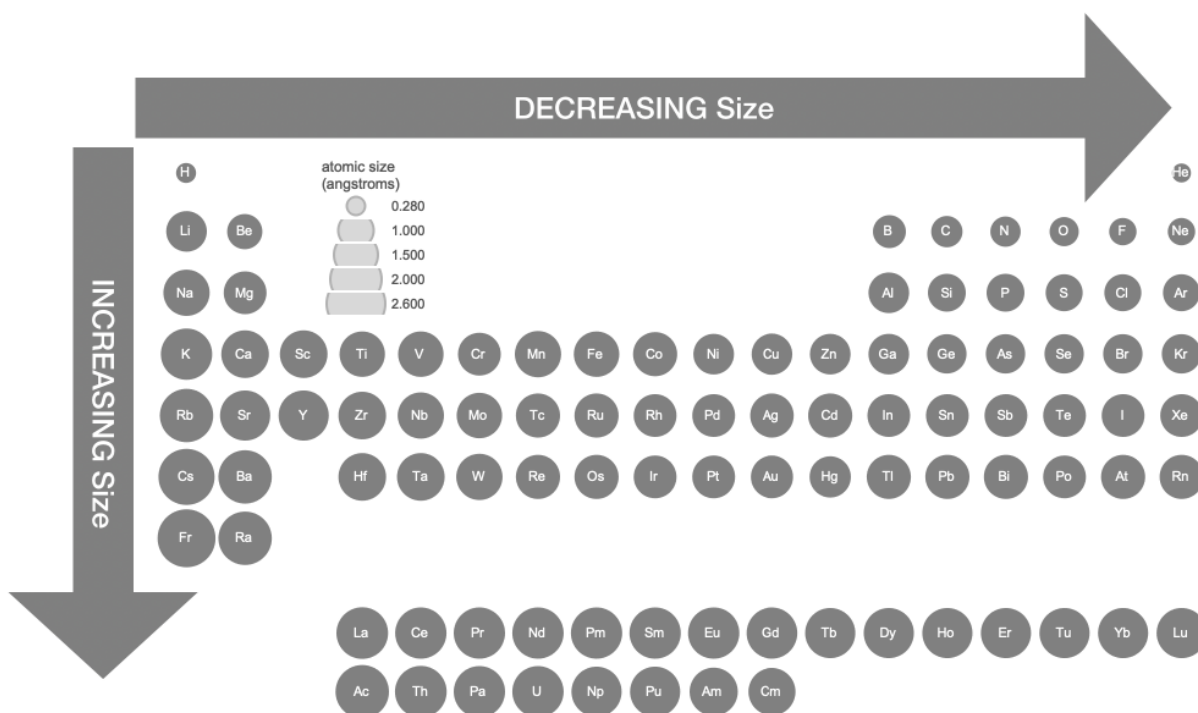


Figure 2.4.4. Atomic Radii Calculated from Crystallographic Data. Data from Cordero, Beatriz, Veronica Gomez, Ana E. Platero-Prats, Marc Reves, Jorge Echeverria, Eduard Cremades, Flavia Barragan, and Santiago Alvarez. “Covalent Radii Revisited.” *Dalton Transactions*, no. 21 (2008): 2832–38. doi:10.1039/b801115j.

Trends in Ionic Radius

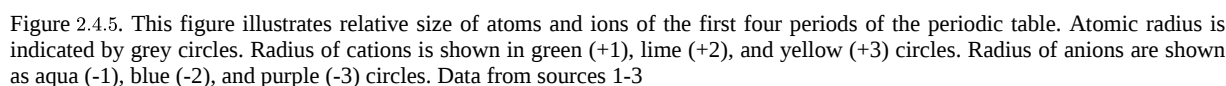
Trends in ionic radius follow general trends in atomic radius for ions of the same charge. Ionic radius varies with charge of the ion (and number of electrons) and the electron configuration (e.g. high spin or low spin).

Cations

Compared to their atoms, cations have the same Z but fewer electrons. Removal of electrons from an atom to form a cation results in a significant increase in effective nuclear charge, resulting in all other electrons being more strongly attracted to the nucleus, and having a lower energy level. The result is a contraction in size from the atom to cation. Figure 2.4.5 visually illustrates the relative size of atoms and some cations of the first four periods; the data is available in tabular format in Figure 2.4.6.

Anions

Compared to their atoms, anions have the same Z but more electrons. Addition of electrons to an atom to form an anion results in a decrease in effective nuclear charge, which corresponds to a decrease in attractive force between the nucleus and electrons. Lower attractive force leads to expansion, where the size of the atom becomes larger in the formation of an anion. Figure 2.4.5 visually illustrates the relative size of atoms and some anions of the first four periods, while the data is available in tabular format in Figure 2.4.6.



H																		He																	
0.31 (atom) 0 (+)																		0.28 (atom)																	
Li																		Ne																	
1.28 (atom) 0.9 (+)		Be																Ar																	
-		-																-																	
Na		Mg																Kr																	
1.66 (atom) 1.16 (+)		1.41 (atom) 0.86 (+)																1.06 (atom)																	
																		Xe																	
K		Ca		Sc		Ti		V		Cr		Mn		Fe		Co		Ni		Cu		Zn		Ga		Ge		As		Se		Br		Kr	
2.03 (atom) 1.52 (+)		1.76 (atom) 1.14 (+)		1.7 (atom) 0.885 (+)		1.6 (atom) 1 (+)		1.53 (atom) 0.93 (+)		1.39 (atom) 0.94 (+)		1.61 (atom) 0.97 (+)		1.52 (atom) 0.92 (+)		1.5 (atom) 0.85 (+)		1.24 (atom) 0.83 (+)		1.32 (atom) 0.91 (+, cm6)		1.22 (atom) 0.88 (+)		1.22 (atom) 0.76 (+)		1.2 (atom) 0.87 (+)		1.19 (atom) 0.72 (+)		1.84 (-)		1.82 (-)		1.16 (atom)	
0.81 (+)		0.41 (+)		0.885 (+)		0.81 (+)		0.93 (+)		0.87 (+)		0.97 (+)		0.92 (+)		0.79 (+)		0.74 (+)		0.87 (+)		0.88 (+)		0.87 (+)		0.87 (+)		0.72 (+)		0.64 (+)		0.73 (+)		0.62 (+)	
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This figure shows radii (in Angstroms) of atoms and ions of the first four periods of the periodic table. Radii from each element are listed from largest to smallest, ionic charge indicated in parentheses (); hs = high spin, ls = low spin, cn6 = coordination number is 6. Data from sources 9-11.

- Swetha Ramireddy (UCD), Bingyao Zheng (UCD), Emily Nguyen (UCD)
- Curated or created by [Kathryn Haas](#)

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CHAPTER OVERVIEW

3: Bonding Theories

This chapter discusses the different theories which can be used to describe bonding in molecules. Some of these theories should be familiar from general or organic chemistry and some may be new to you this semester. Not every theory is applicable in every situation so, so we will look at cases where these theories are successful and cases where they fail (can't explain or reproduce experimental observations).

Learning Objectives

- Apply Lewis and VSEPR theories to determine molecular geometry and bond angles
- Identify polar and non-polar molecules
- Model bonding and orbital hybridization with s and p atomic orbitals
- Illustrate how atomic orbitals can overlap to form sigma, pi, delta, or non-bonding interactions
- Construct a qualitative molecular orbital diagram for a homo or heteronuclear diatomic molecule

The thumbnail image compares the Lewis dot structure with the sigma antibonding molecular orbital in HCl. The molecular orbital was calculated and rendered with WebMO.

[3.1: Types of Bonding](#)

[3.2: Electronegativity Values](#)

[3.2A: Pauling Electronegativity Values](#)

[3.2B: Mulliken Electronegativity Values](#)

[3.2C: Allred-Rochow Electronegativity Values](#)

[3.3: Lewis Electron-Dot Diagrams](#)

[3.3A: Formal Charge](#)

[3.3B: Resonance](#)

[3.3C: Expanded Octets](#)

[3.3D: Limitation of Lewis Theory](#)

[3.4: Valence-Shell Electron-Repulsion Theory](#)

[3.5: Molecular Polarity](#)

[3.6: Valence Bond Theory](#)

[3.7: Molecular Orbital Theory](#)

[3.7A: Orbital Overlap](#)

[3.7B: Homonuclear Diatomic Molecules](#)

[3.7C: Heteronuclear Diatomic Molecules](#)

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3.1: Types of Bonding

In general chemistry we learned that bonding between atoms can be classified as a range of possible bonding between **ionic bonds** (fully charge transfer) and **covalent bonds** (fully shared electrons). When two atoms of slightly differing electronegativities come together to form a covalent bond, one atom attracts the electrons more than the other; this is called a polar covalent bond. However, simple “ionic” and “covalent” bonding are idealized concepts and most bonds exist on a two-dimensional continuum described by the van Arkel-Ketelaar Triangle (Figure 3.1.1).

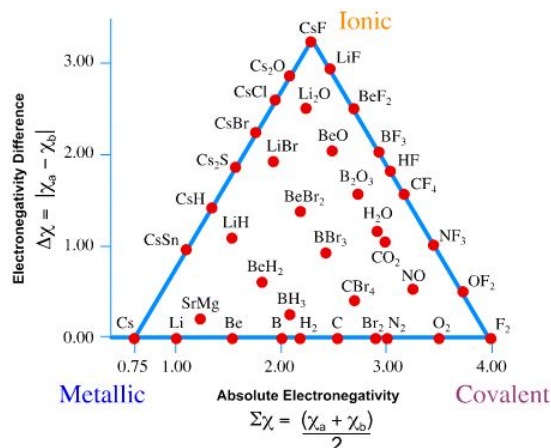


Figure 3.1.1: van Arkel-Ketelaar Triangle plots the difference in electronegativity ($\Delta\chi$) and the average electronegativity in a bond ($\Sigma\chi$). the top region is where bonds are mostly ionic, the lower left region is where bonding is metallic, and the lower right region is where the bonding is covalent.

Bond triangles or **van Arkel-Ketelaar** triangles (named after Anton Eduard van Arkel and J. A. A. Ketelaar) are triangles used for showing different compounds in varying degrees of ionic, metallic and covalent bonding. In 1941 van Arkel recognized three extreme materials and associated bonding types. Using 36 main group elements, such as metals, metalloids and non-metals, he placed ionic, metallic and covalent bonds on the corners of an equilateral triangle, as well as suggested intermediate species. The bond triangle shows that chemical bonds are not just particular bonds of a specific type. Rather, bond types are interconnected and different compounds have varying degrees of different bonding character (for example, polar covalent bonds).



Video 3.1.1: What is the van Arkel-Ketelaar Triangle of Bonding?

Using electronegativity - two compound average electronegativity on x-axis of Figure 3.1.1.

$$\Sigma\chi = \frac{\chi_A + \chi_B}{2} \quad (3.1.1)$$

and electronegativity difference on y-axis,

$$\Delta\chi = |\chi_A - \chi_B| \quad (3.1.2)$$

we can rate the dominant bond between the compounds. On the right side of Figure 3.1.1 (from ionic to covalent) should be compounds with varying difference in electronegativity. The compounds with equal electronegativity, such as Cl_2 (chlorine) are placed in the covalent corner, while the ionic corner has compounds with large electronegativity difference, such as NaCl (table salt). The bottom side (from metallic to covalent) contains compounds with varying degree of directionality in the bond. At one extreme is metallic bonds with delocalized bonding and at the other are covalent bonds in which the orbitals overlap in a particular direction. The left side (from ionic to metallic) is meant for delocalized bonds with varying electronegativity difference.

The Three Extremes in bonding

In general:

- Metallic bonds have low $\Delta\chi$ **and** low average $\sum\chi$.
- Ionic bonds have moderate-to-high $\Delta\chi$ and moderate values of average $\sum\chi$.
- Covalent bonds have moderate to high average $\sum\chi$ and can exist with moderately low $\Delta\chi$.

Example 3.1.1

Use the tables of electronegativities and Figure 3.1.1 to estimate the following values

- difference in electronegativity ($\Delta\chi$)
- average electronegativity in a bond ($\sum\chi$)
- percent ionic character
- likely bond type

for the selected compounds:

- AsH (e.g., in arsine AsH_3)
- SrLi
- KF.

Solution

a: AsH

- The electronegativity of As is 2.18
- The electronegativity of H is 2.22

Using Equations 3.1.1 and 3.1.2:

$$\begin{aligned}\sum\chi &= \frac{\chi_A + \chi_B}{2} \\ &= \frac{2.18 + 2.22}{2} \\ &= 2.2\end{aligned}$$

$$\begin{aligned}\Delta\chi &= \chi_A - \chi_B \\ &= 2.18 - 2.22 \\ &= 0.04\end{aligned}$$

- From Figure 3.1.1, the bond is fairly nonpolar and has a low ionic character (10% or less)
- The bonding is in the middle of a covalent bond and a metallic bond

b: SrLi

- The electronegativity of Sr is 0.95
- The electronegativity of Li is 0.98

Using Equations 3.1.1 and 3.1.2:

$$\begin{aligned}\sum \chi &= \frac{\chi_A + \chi_B}{2} \\ &= \frac{0.95 + 0.98}{2} \\ &= 0.965\end{aligned}$$

$$\begin{aligned}\Delta\chi &= \chi_A - \chi_B \\ &= 0.98 - 0.95 \\ &= 0.025\end{aligned}$$

- From Figure 3.1.1, the bond is fairly nonpolar and has a low ionic character (~3% or less)
- The bonding is likely metallic.

c: KF

- The electronegativity of K is 0.82
- The electronegativity of F is 3.98

Using Equations 3.1.1 and 3.1.2:

$$\begin{aligned}\sum \chi &= \frac{\chi_A + \chi_B}{2} \\ &= \frac{0.82 + 3.98}{2} \\ &= 2.4\end{aligned}$$

$$\begin{aligned}\Delta\chi &= \chi_A - \chi_B \\ &= |0.82 - 3.98| \\ &= 3.16\end{aligned}$$

- From Figure 3.1.1, the bond is fairly polar and has a high ionic character (~75%)
- The bonding is likely ionic.

Exercise 3.1.1

Contrast the bonding of NaCl and silicon tetrafluoride.

Answer

NaCl is an ionic crystal structure, and an electrolyte when dissolved in water; $\Delta\chi = 1.58$, average $\sum \chi = 1.79$, while silicon tetrafluoride is covalent (molecular, non-polar gas; $\Delta\chi = 2.08$, average $\sum \chi = 2.94$).

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